

MACHINE LEARNING CONCEPTS AND APPLICATIONS IN PYTHON (PART 3)

DR. SAMRAT MONDAL

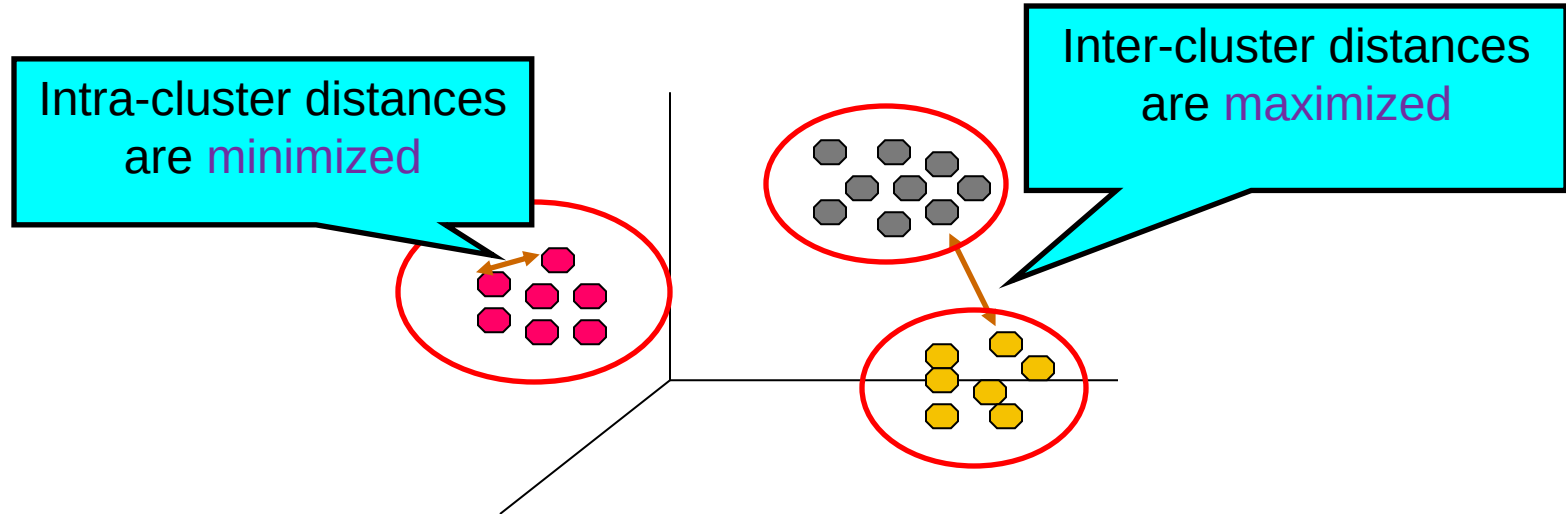
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

INDIAN INSTITUTE OF TECHNOLOGY PATNA

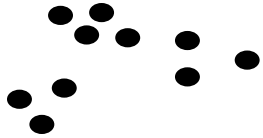
CLUSTERING

WHAT IS CLUSTER ANALYSIS?

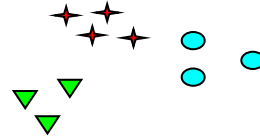
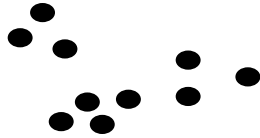
- Given a set of objects, place them in groups such that the objects in a group are similar (or related) to one another and different from (or unrelated to) the objects in other groups
- An Unsupervised Learning



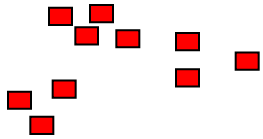
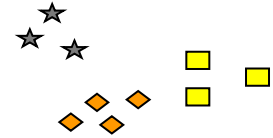
NOTION OF A CLUSTER CAN BE AMBIGUOUS



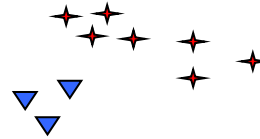
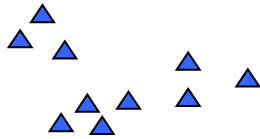
How many clusters?



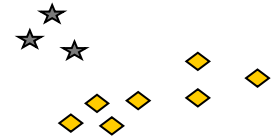
Six Clusters



Two Clusters



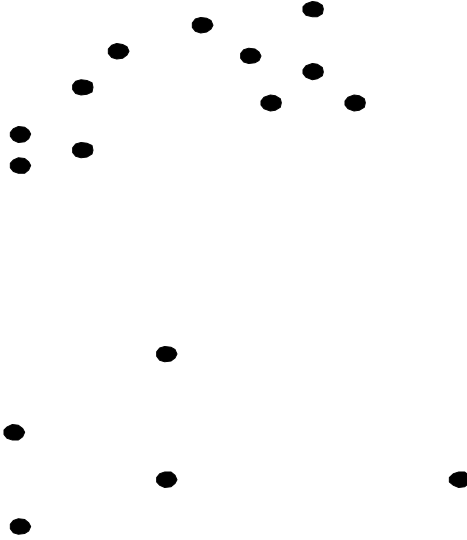
Four Clusters



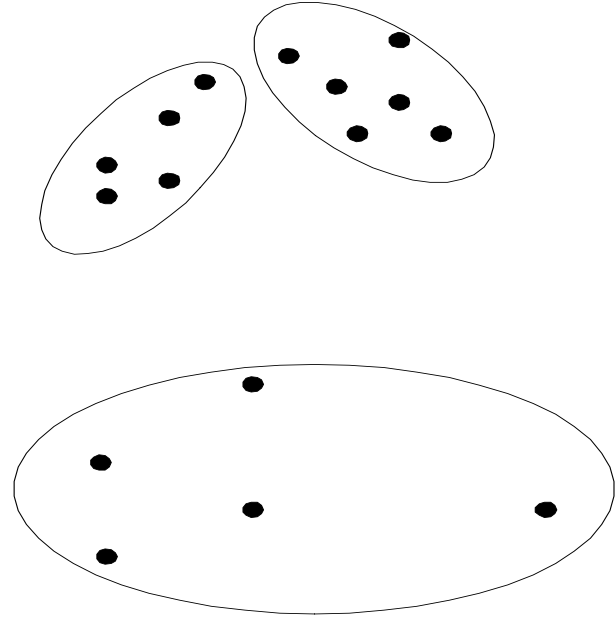
TYPES OF CLUSTERING

- A **clustering** represents the division of data into distinct groups, called **clusters**, where each cluster contains items that share similar characteristics or patterns
- Important distinction between **hierarchical** and **partitional** sets of clusters
 - **Partitional Clustering**
 - A division of data objects into non-overlapping subsets (clusters)
 - **Hierarchical clustering**
 - A set of nested clusters organized as a hierarchical tree

PARTITIONAL CLUSTERING

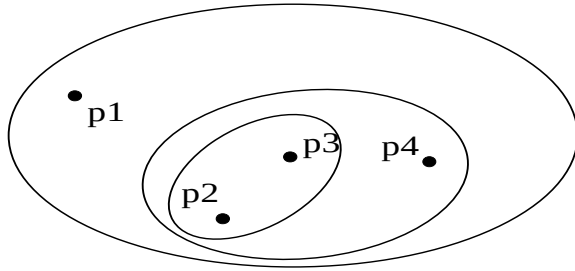


Original Points

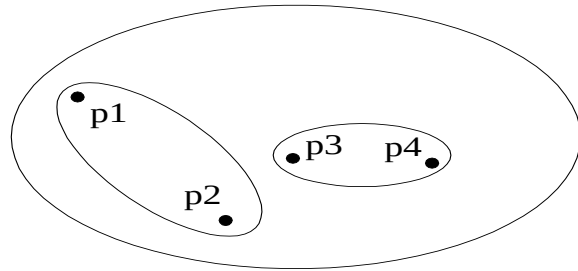


A Partitional Clustering

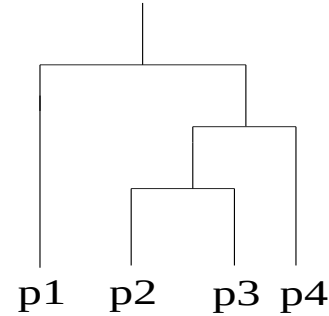
HIERARCHICAL CLUSTERING



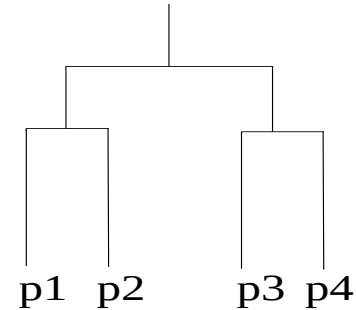
Hierarchical Clustering 1



Hierarchical Clustering 2



Dendrogram 1



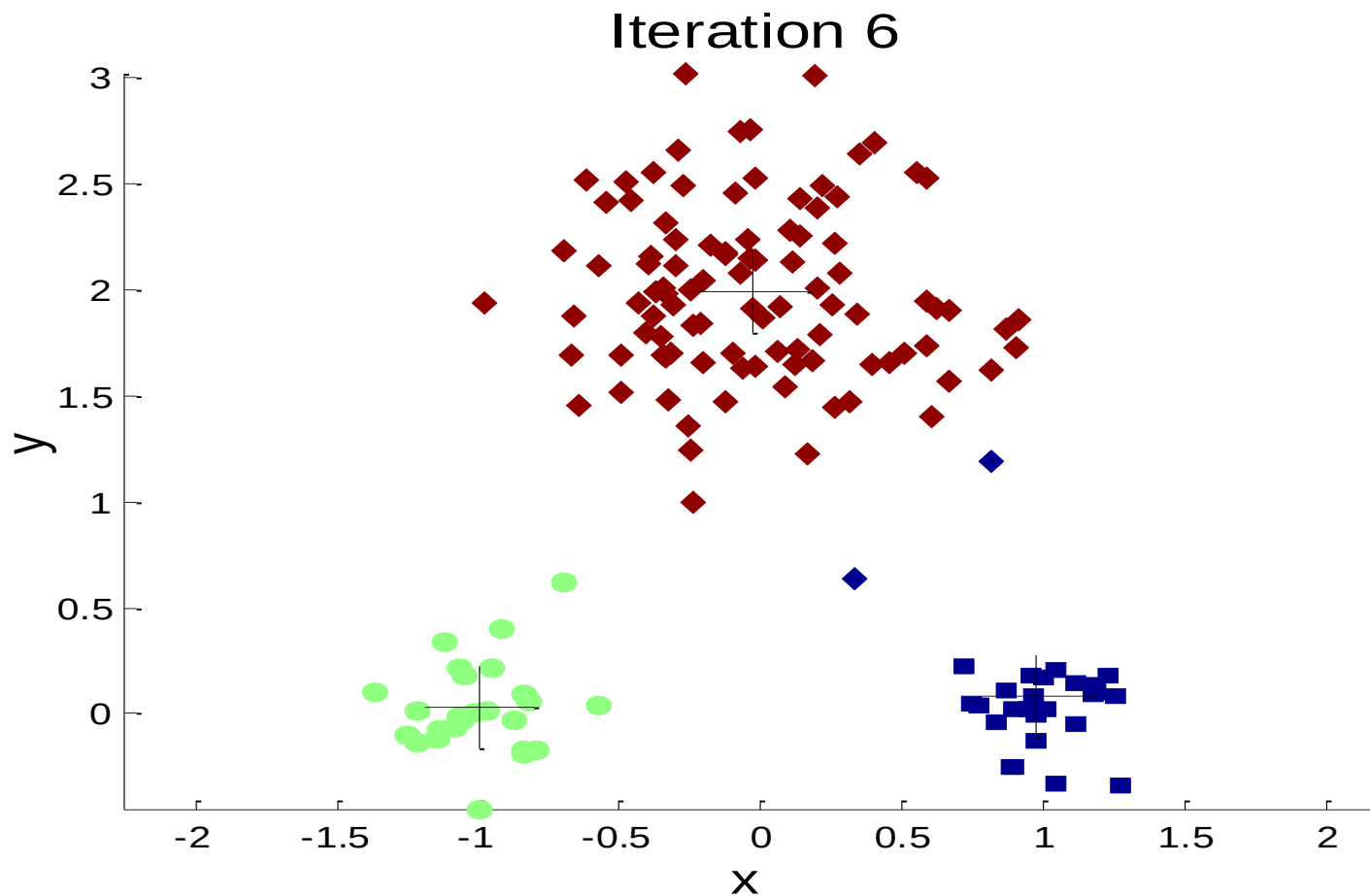
Dendrogram 2

K-MEANS CLUSTERING

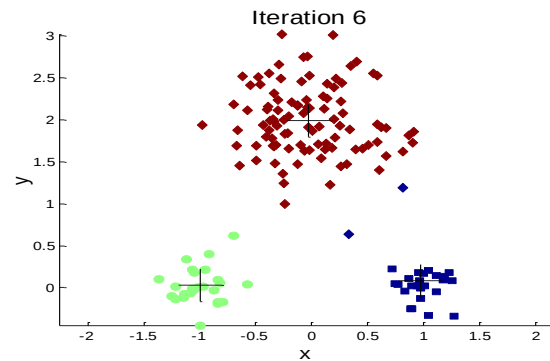
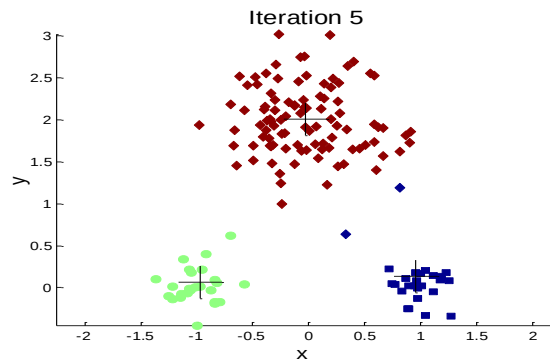
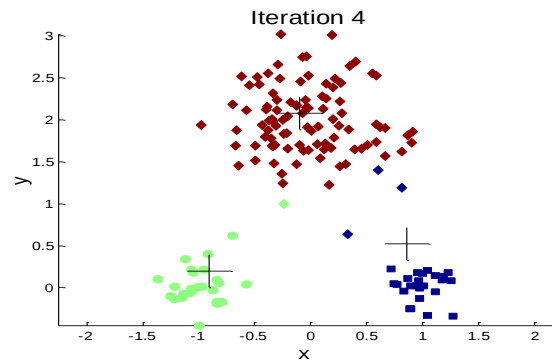
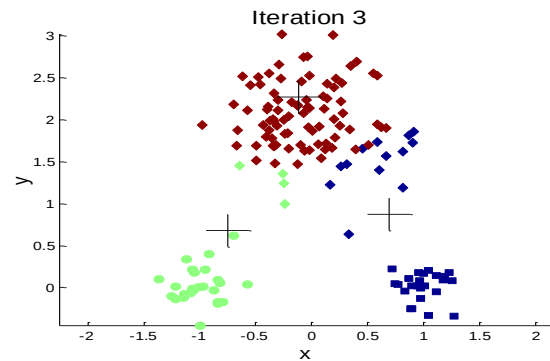
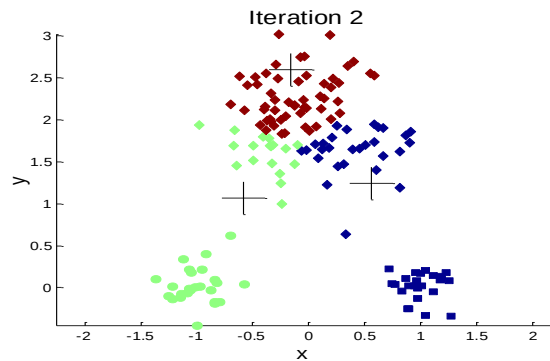
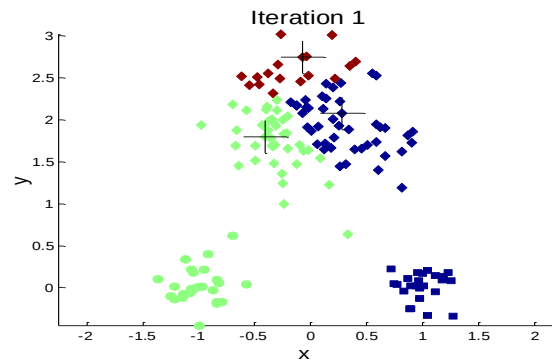
- **Partitional** clustering approach
- Number of clusters, K , must be specified
- Each cluster is associated with a **centroid** (center point)
- Each point is assigned to the cluster with the closest centroid
- The basic algorithm is very simple

-
- 1: Select K points as the initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning all points to the closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** The centroids don't change
-

EXAMPLE OF K-MEANS CLUSTERING



EXAMPLE OF K-MEANS CLUSTERING



K-MEANS CLUSTERING – DETAILS

- Simple **iterative** algorithm.
 - Choose initial centroids;
 - repeat {assign each point to a nearest centroid; re-compute cluster centroids}
 - until centroids stop changing.
- **Initial centroids** are often chosen randomly.
 - Clusters produced can vary from one run to another
- The **centroid** is (typically) the mean of the points in the cluster, but other definitions are possible.

K-MEANS CLUSTERING – DETAILS

- K-means will converge for common proximity measures with appropriately defined centroid
- Most of the convergence happens in the first few iterations.
 - Often the stopping condition is changed to 'Until relatively few points change clusters'
- Complexity is $O(n * K * I * d)$
 - n = number of points, K = number of clusters,
 I = number of iterations, d = number of attributes

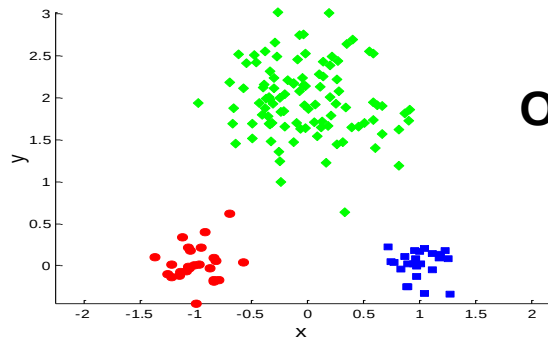
K-MEANS OBJECTIVE FUNCTION

- A common **objective function** (used with Euclidean distance measure) is **Sum of Squared Error (SSE)**
 - For each point, the **error** is the distance to the nearest cluster center
 - To get SSE, we square these errors and sum them.

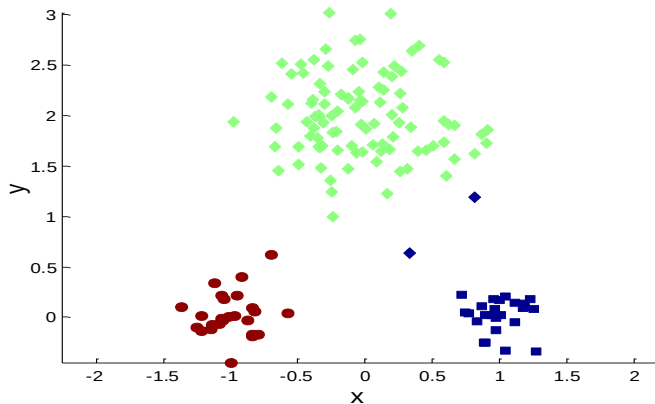
$$SSE = \sum_{i=1}^k \sum_{x \in c_i} dist(m_i, x)^2$$

- x is a data point in cluster C_i and m_i is the centroid (mean) for cluster C_i
- SSE improves in each iteration of K-means until it reaches a local or global minima.

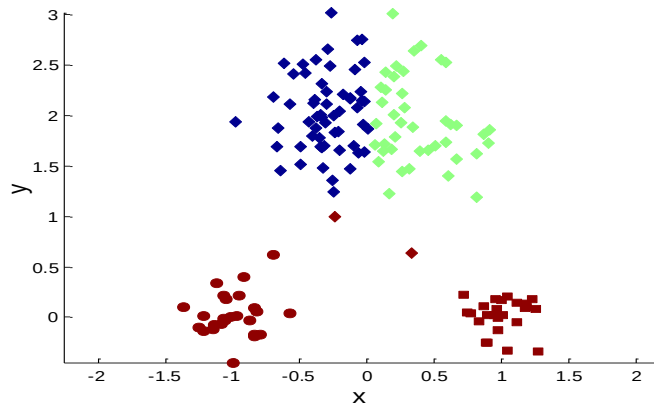
TWO DIFFERENT K-MEANS CLUSTERINGS



Original Points

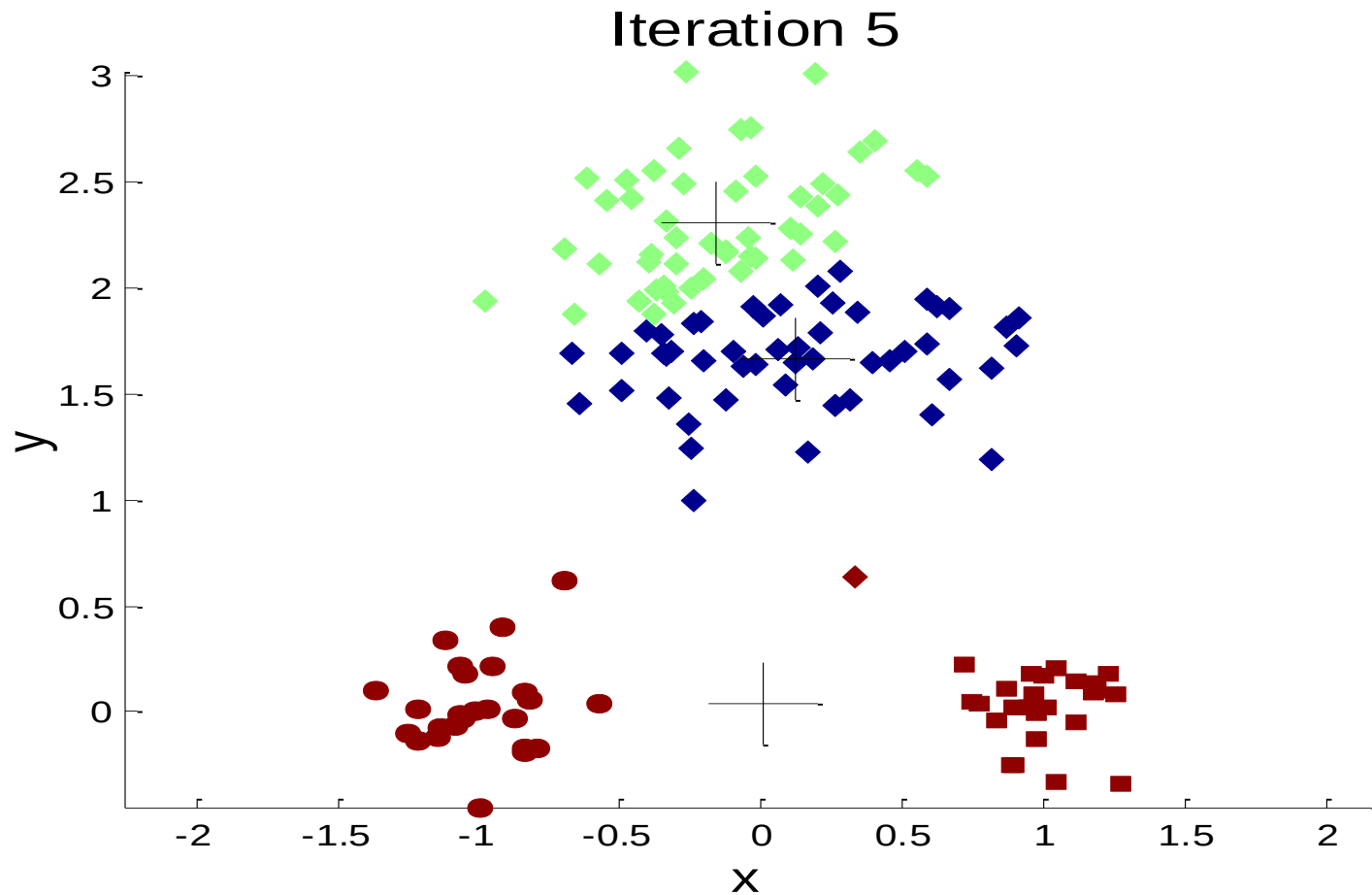


Optimal Clustering

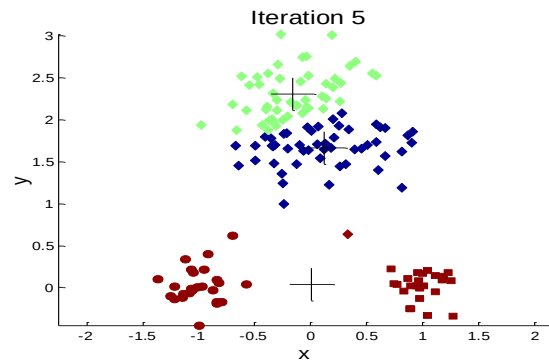
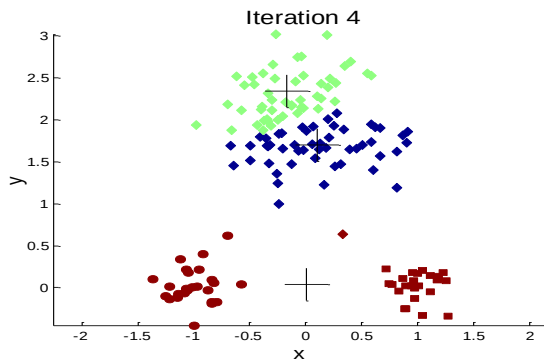
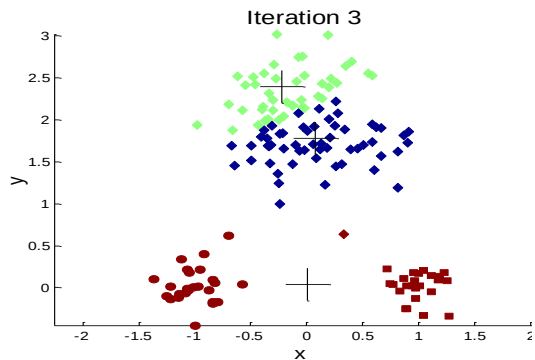
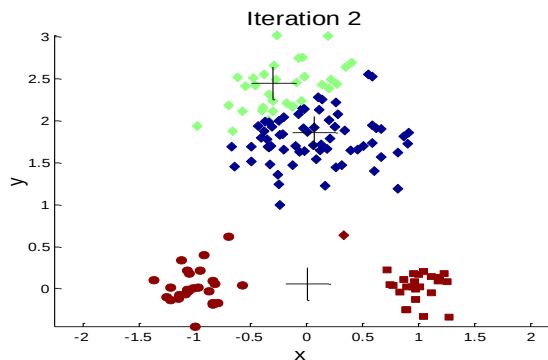
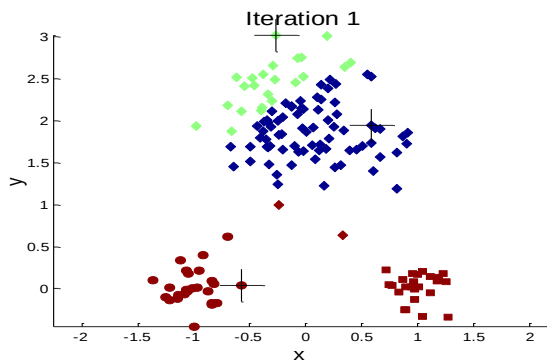


Sub-optimal Clustering

IMPORTANCE OF CHOOSING INITIAL CENTROIDS

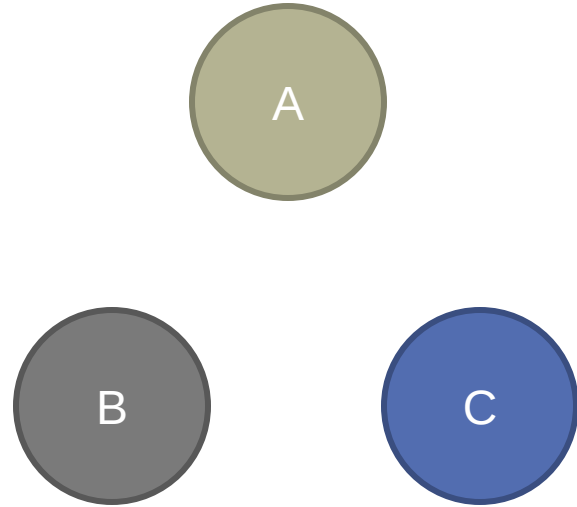


IMPORTANCE OF CHOOSING INITIAL CENTROIDS

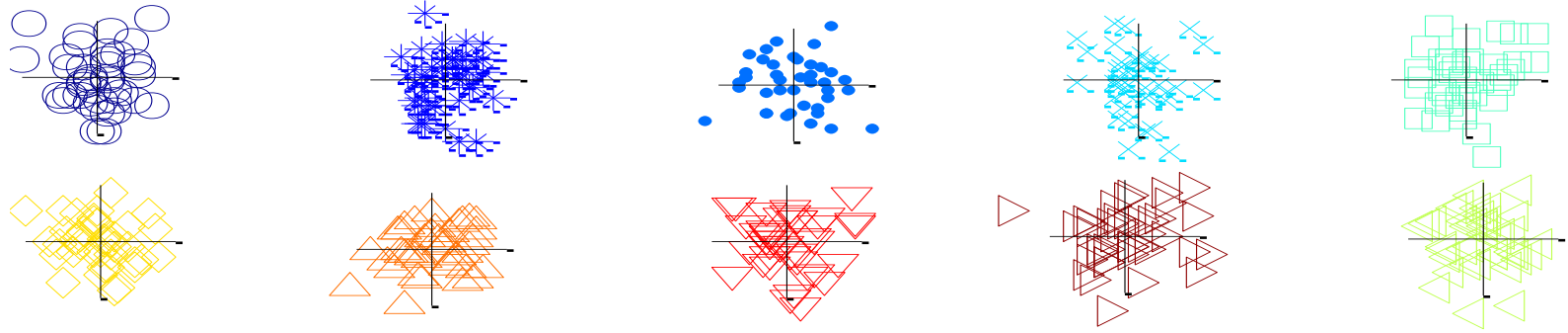


IMPORTANCE OF CHOOSING INITIAL CENTROIDS

Depending on the choice of initial centroids, B and C may get merged or remain separate

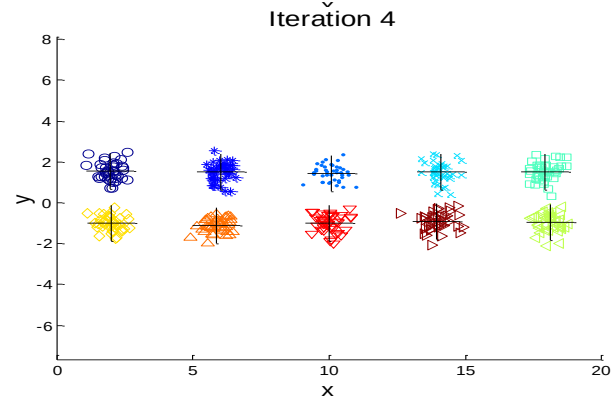
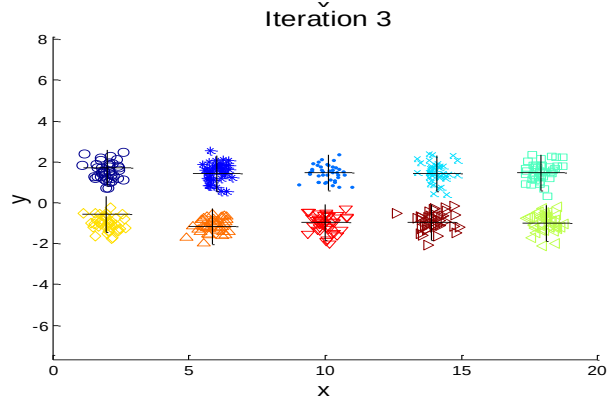
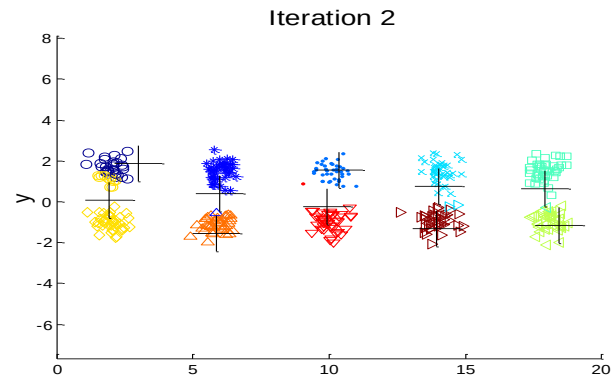
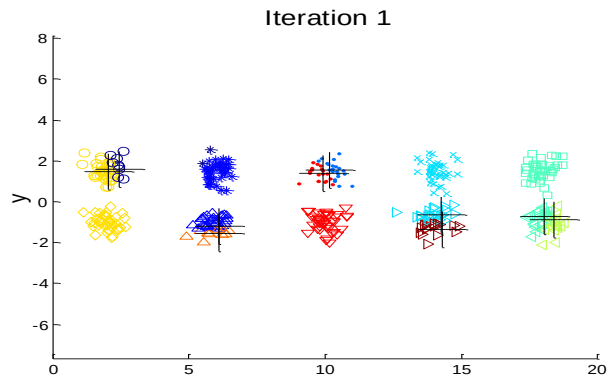


10 CLUSTERS EXAMPLE



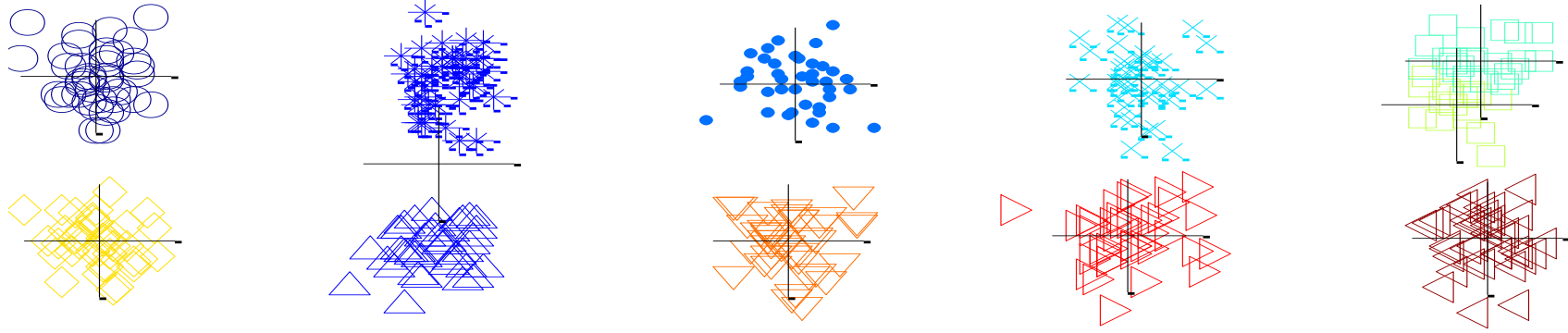
Starting with two initial centroids in one cluster of each pair of clusters

10 CLUSTERS EXAMPLE



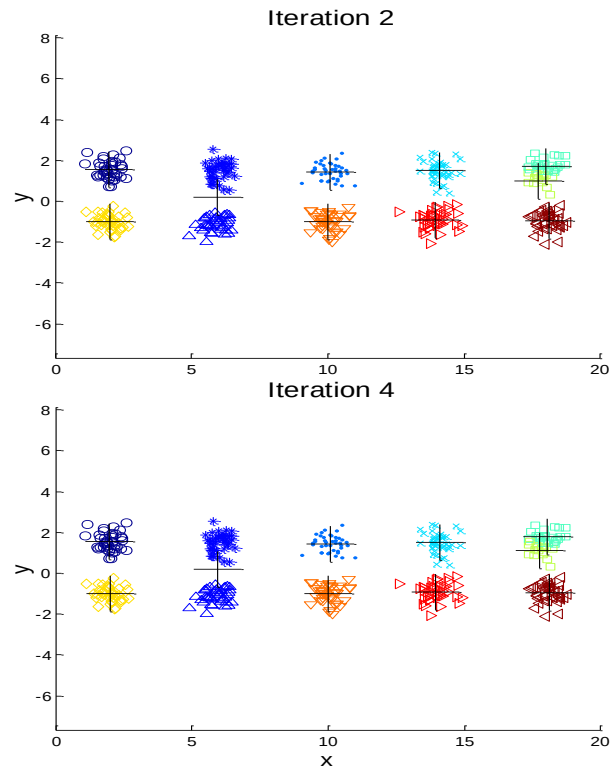
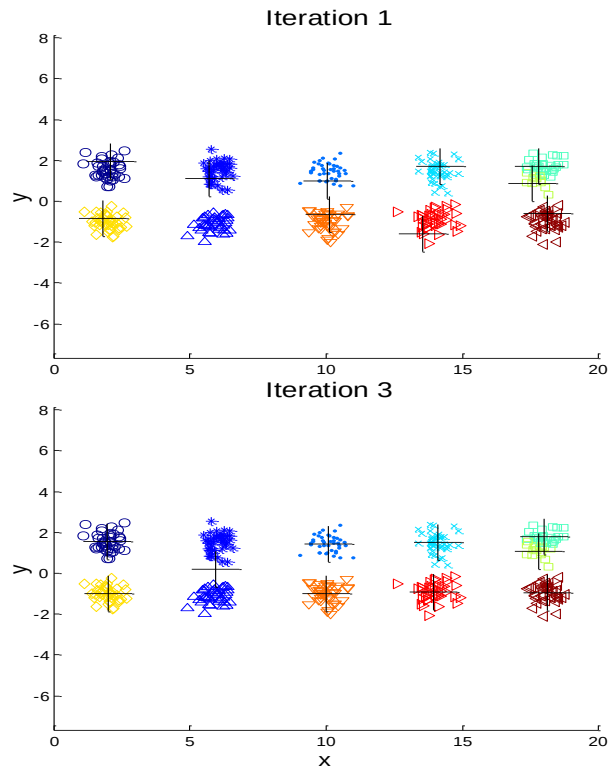
Starting with two initial centroids in one cluster of each pair of clusters

10 CLUSTERS EXAMPLE



Starting with some pairs of clusters having three initial centroids, while other have only one.

10 CLUSTERS EXAMPLE

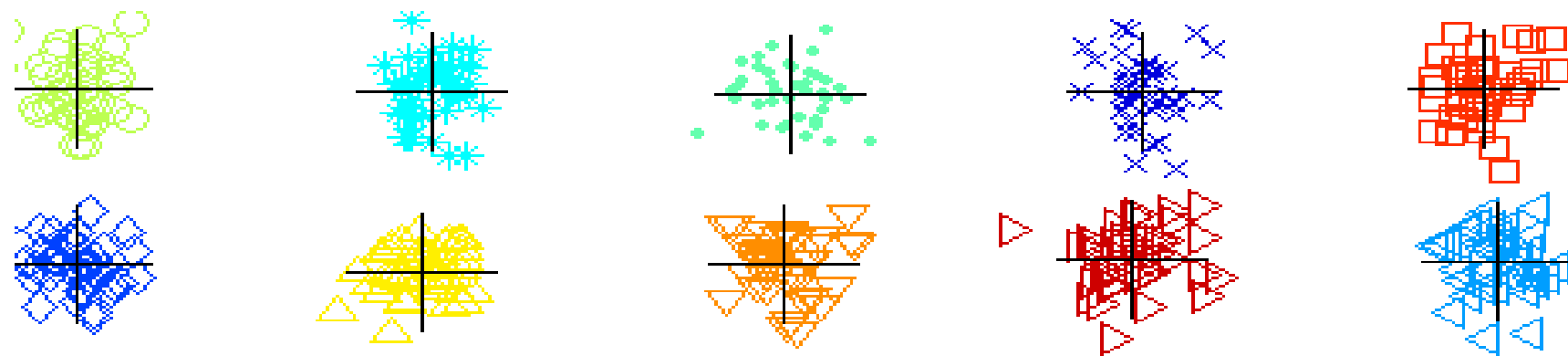


Starting with some pairs of clusters having three initial centroids, while other have only one.

SOLUTIONS TO INITIAL CENTROIDS PROBLEM

- Multiple runs
 - Helps, but probability is not on your side
- Use some strategy to select the k initial centroids and then select among these initial centroids
 - Select most widely separated
 - K-means++ is a robust way of doing this selection
 - Use hierarchical clustering to determine initial centroids
 - Bisecting K-means: Not as susceptible to initialization issues

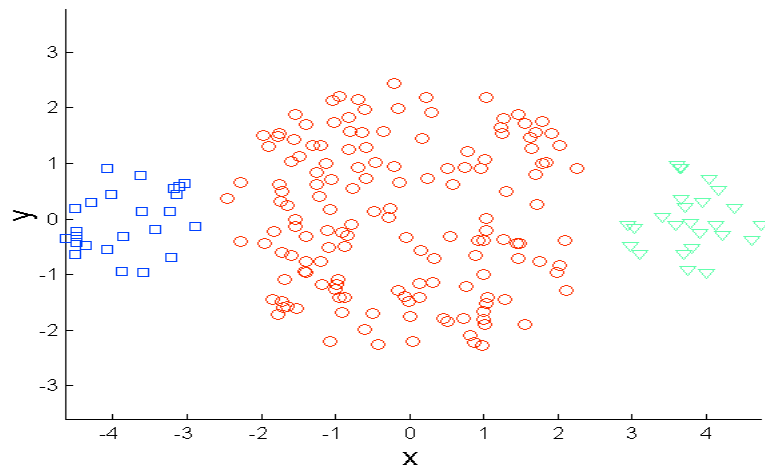
BISECTING K-MEANS EXAMPLE



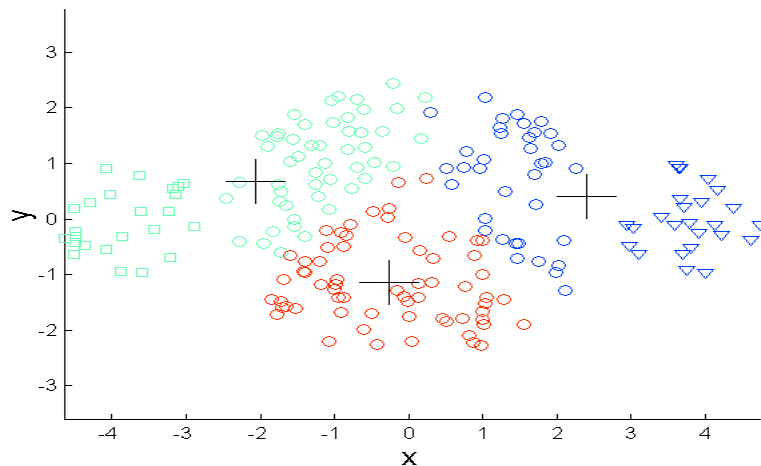
LIMITATIONS OF K-MEANS

- K-means has problems when clusters are of differing
 - Sizes
 - Densities
 - Non-globular shapes
- K-means has problems when the data contains outliers.
 - One possible solution is to remove outliers before clustering

LIMITATIONS OF K-MEANS: DIFFERING SIZES

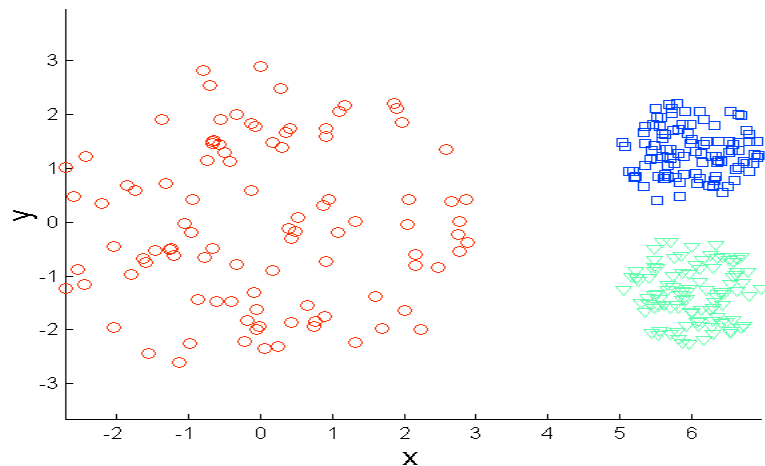


Original Points

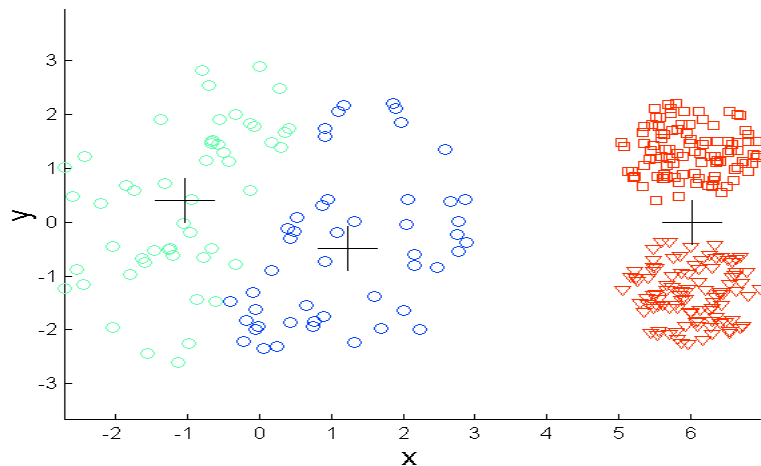


K-means (3 Clusters)

LIMITATIONS OF K-MEANS: DIFFERING DENSITY

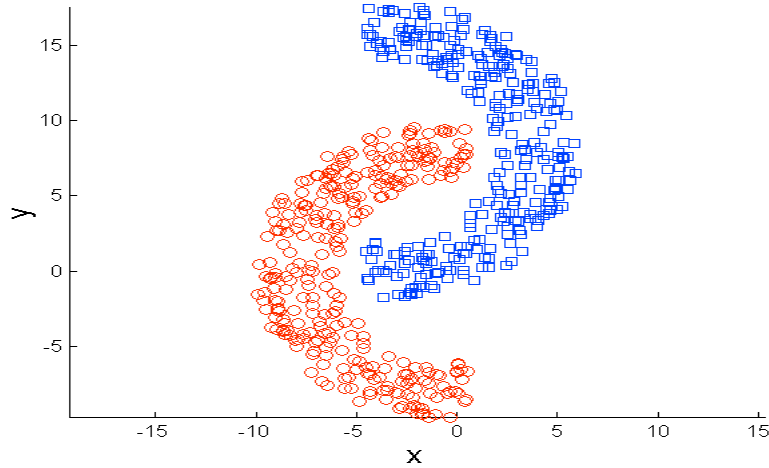


Original Points

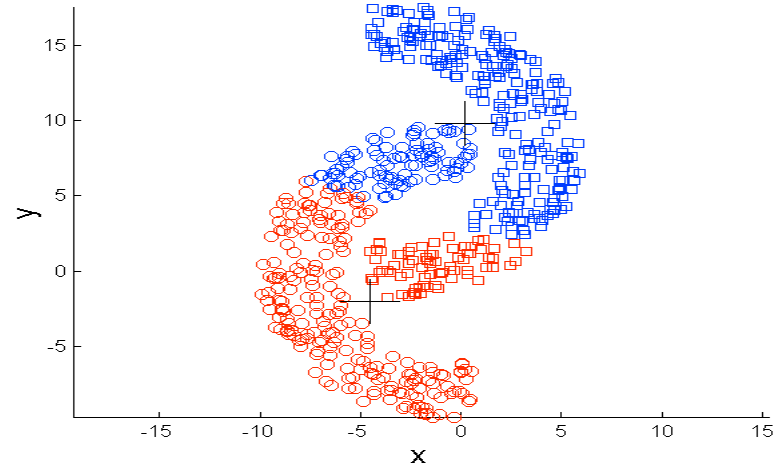


K-means (3 Clusters)

LIMITATIONS OF K-MEANS: NON-GLOBULAR SHAPES

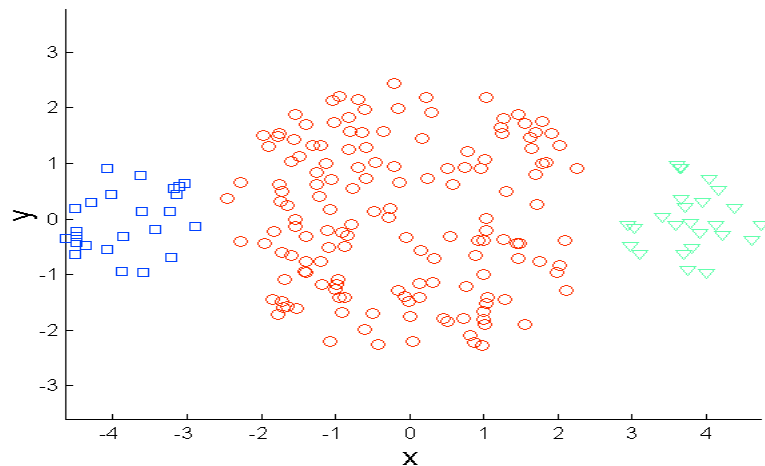


Original Points

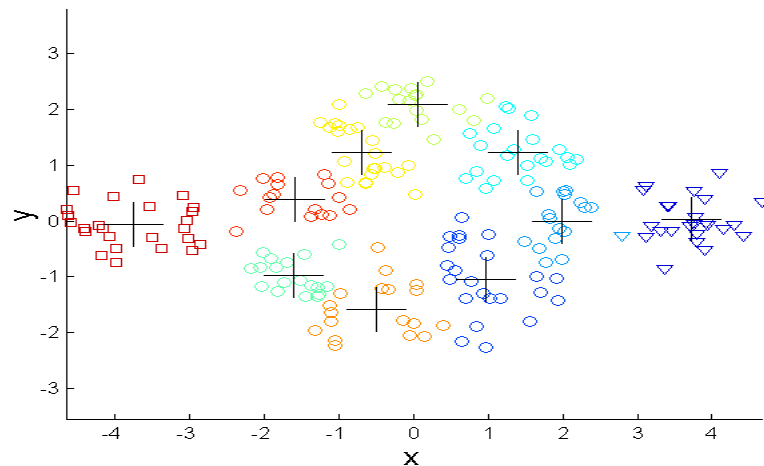


K-means (2 Clusters)

OVERCOMING K-MEANS LIMITATIONS



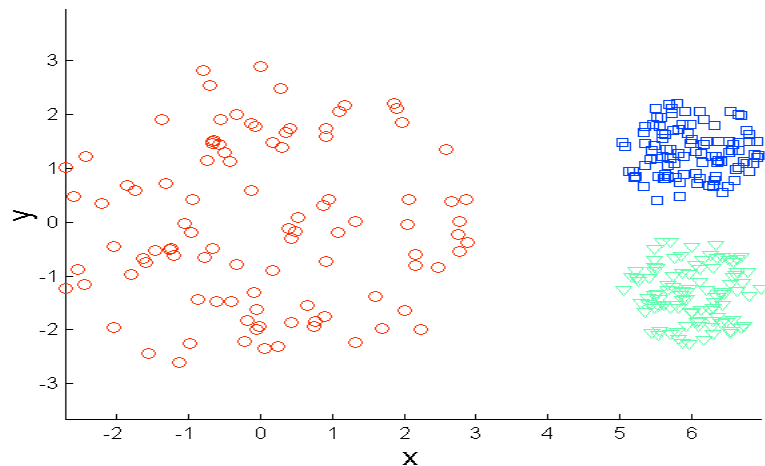
Original Points



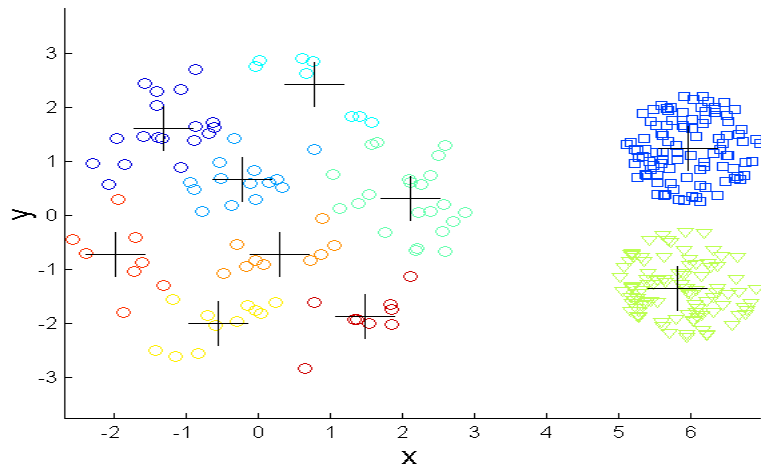
K-means Clusters

One solution is to find a large number of clusters such that each of them represents a part of a natural cluster. But these small clusters need to be put together in a post-processing step.

OVERCOMING K-MEANS LIMITATIONS



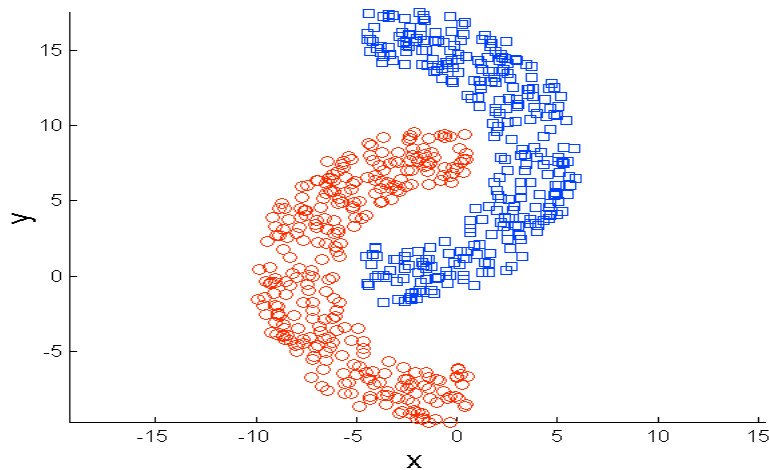
Original Points



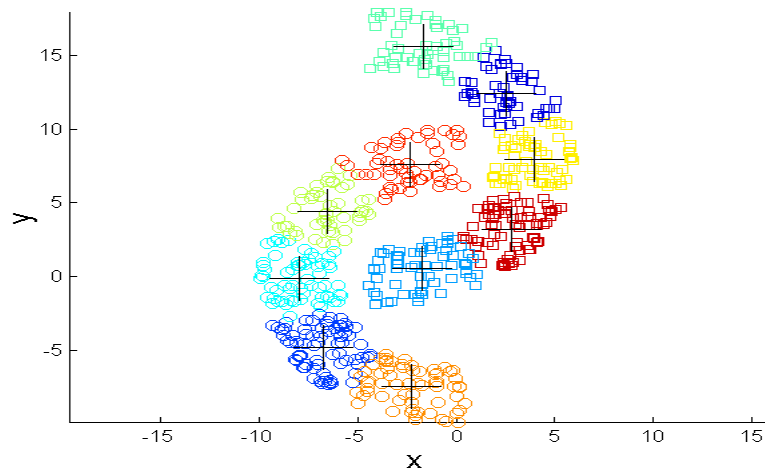
K-means Clusters

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OVERCOMING K-MEANS LIMITATIONS



Original Points



K-means Clusters

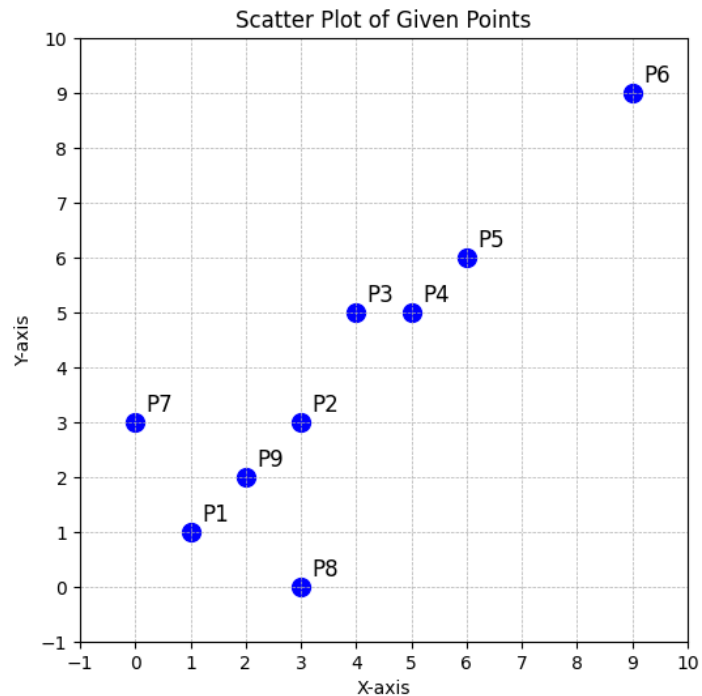
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KMEANS: EXERCISE

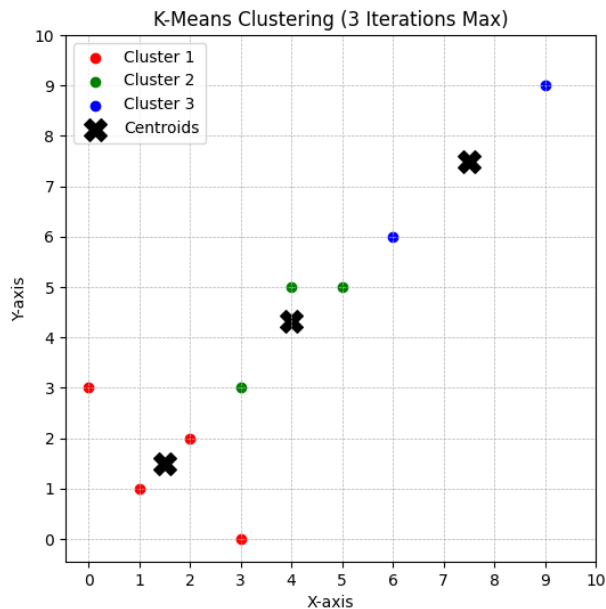
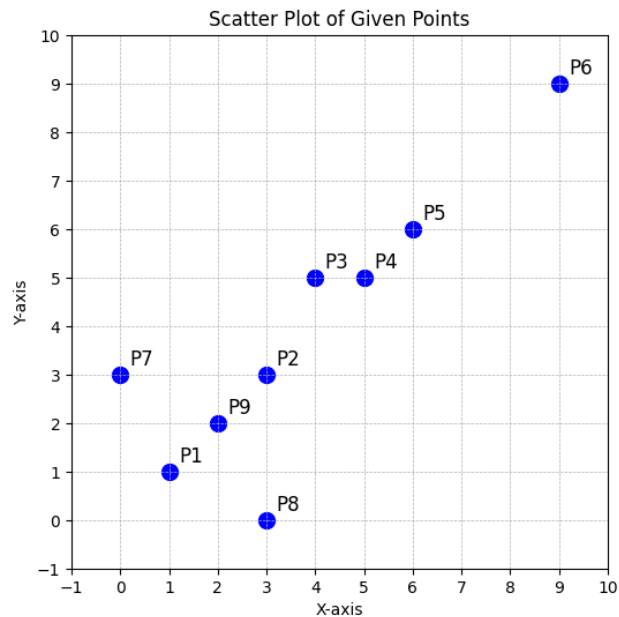
Points	X	Y
P1	1	1
P2	3	3
P3	4	5
P4	5	5
P5	6	6
P6	9	9
P7	0	3
P8	3	0
P9	2	2

K=3

Initial
Centroids:
(1,1)
(3,3)
(5,5)



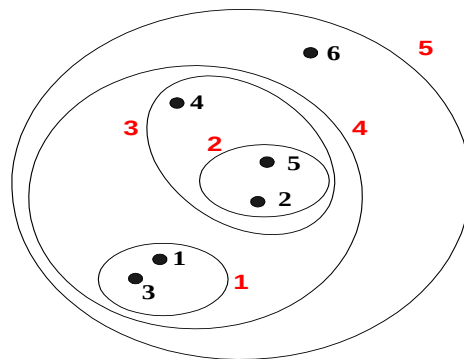
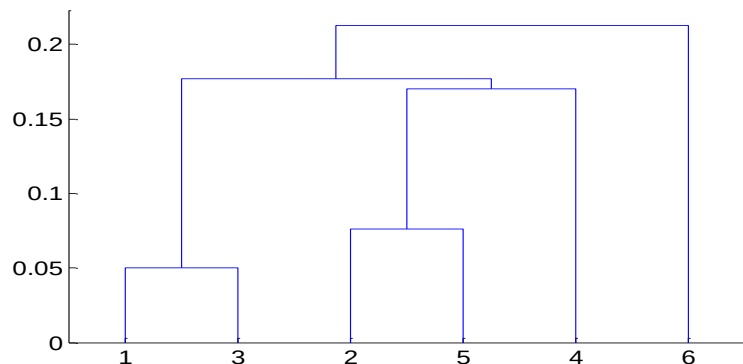
RESULT



Points	X	Y
P1	1	1
P2	3	3
P3	4	5
P4	5	5
P5	6	6
P6	9	9
P7	0	3
P8	3	0
P9	2	2

HIERARCHICAL CLUSTERING (HC)

- Produces a set of nested clusters organized as a **hierarchical tree**
- Can be visualized as a **dendrogram**
 - A tree like diagram that records the sequences of merges or splits



STRENGTHS OF HC

- Do not have to assume any particular number of clusters
 - Any **desired number of clusters** can be obtained by ‘cutting’ the dendrogram at the proper level
- They may correspond to meaningful **taxonomies**
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

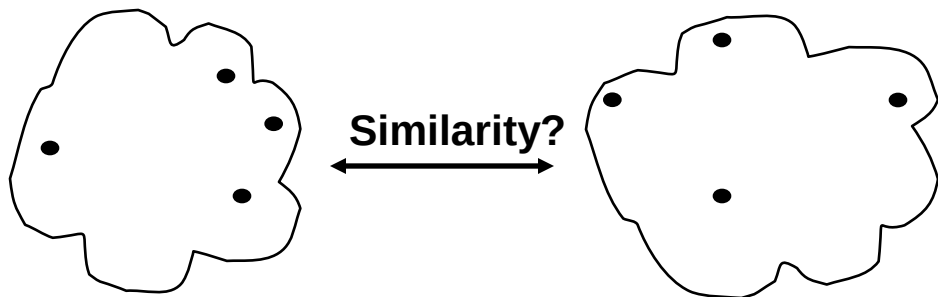
HC: TYPES

- **Agglomerative:**
 - Start with the points as individual clusters
 - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
- **Divisive:**
 - Start with one, all-inclusive cluster
 - At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
 - Merge or split one cluster at a time

AGGLOMERATIVE CLUSTERING ALGORITHM

- **Key Idea:** Successively merge closest clusters
- Basic algorithm
 1. Compute the **proximity matrix**
 2. Let each data point be a cluster
 3. Repeat
 4. Merge the two closest clusters
 5. Update the **proximity matrix**
 6. Until only a single cluster remains
- Key operation is the computation of the proximity of two clusters
 - Different approaches in defining the distance between clusters distinguish the different algorithms

HOW TO DEFINE INTER-CLUSTER DISTANCE

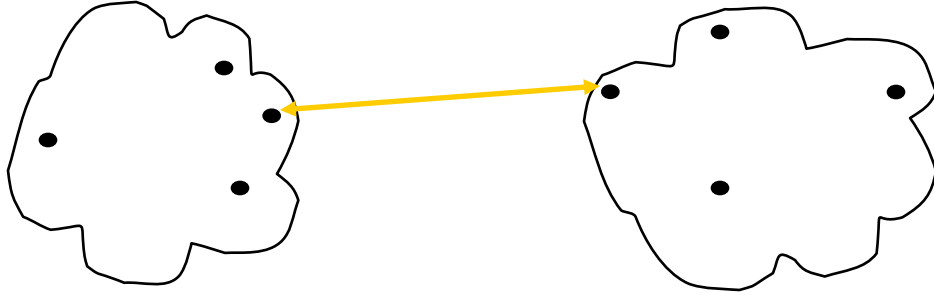


- MIN
- MAX
- Group Average
- Distance Between Centroids
- Other methods driven by an objective function
 - Ward's Method uses squared error

	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

Proximity Matrix

HOW TO DEFINE INTER-CLUSTER SIMILARITY

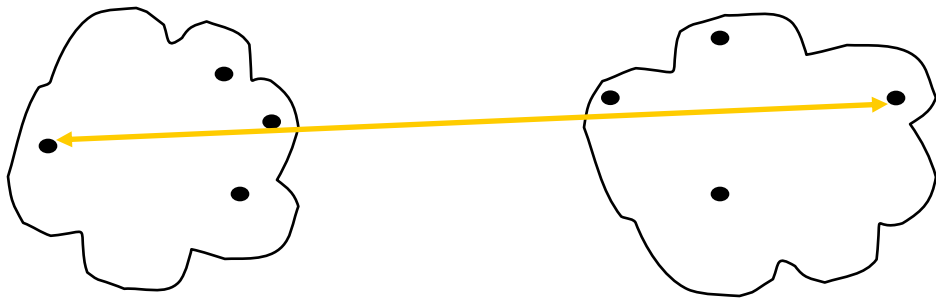


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Proximity Matrix

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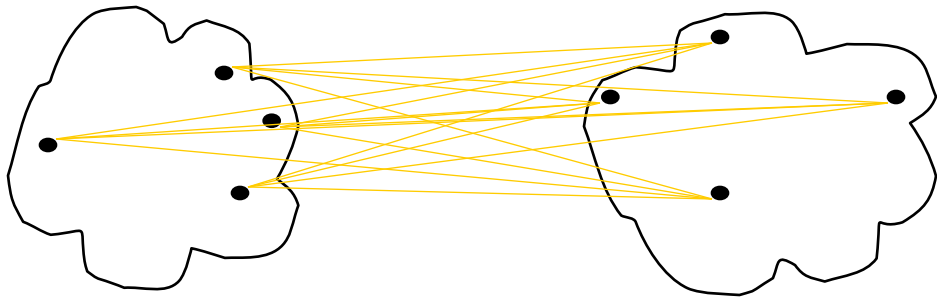


- MIN
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Proximity Matrix

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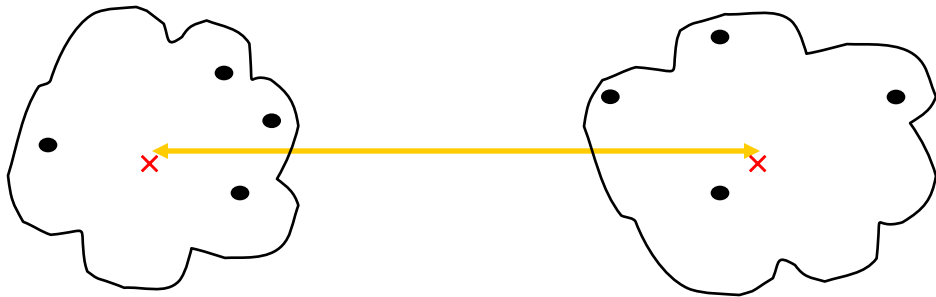


- MIN
- MAX
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	p1	p2	p3	p4	p5	...
p1						
p2						
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Proximity Matrix

HOW TO DEFINE INTER-CLUSTER SIMILARITY



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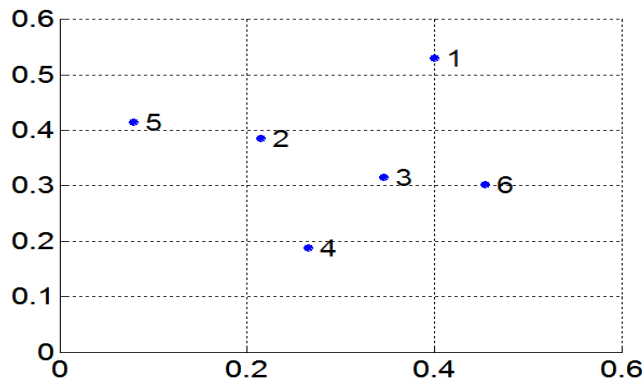
	p1	p2	p3	p4	p5	...
p1						
p2						
p3						
p4						
p5						

Proximity Matrix

MIN OR SINGLE LINK

- Proximity of two clusters is based on the two closest points in the different clusters
 - Determined by one pair of points, i.e., by one link in the proximity graph

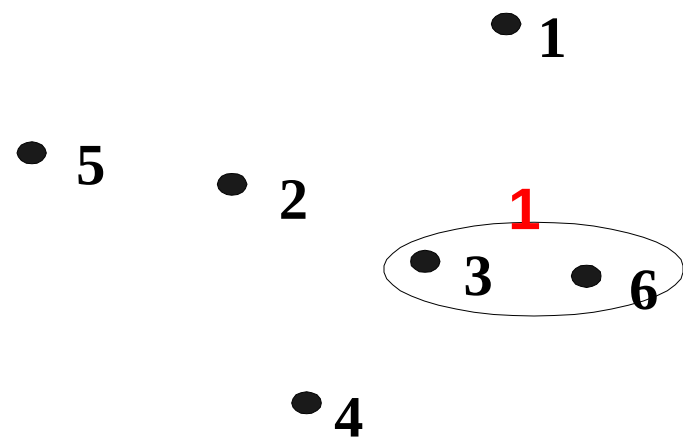
Example:



Distance Matrix:

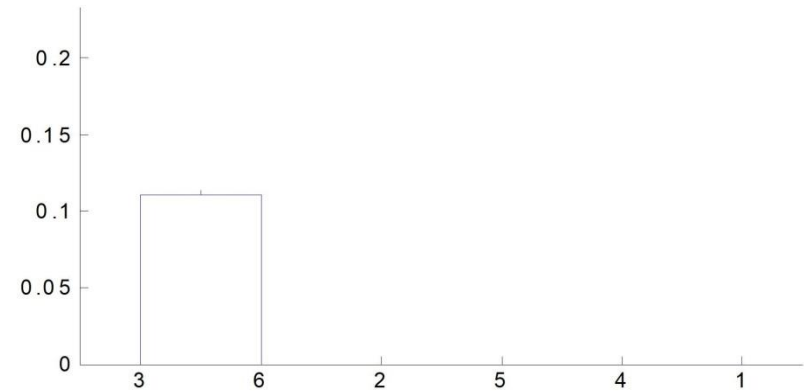
	p_1	p_2	p_3	p_4	p_5	p_6
p_1	0	0.24	0.22	0.37	0.34	0.23
p_2	0.24	0	0.15	0.20	0.14	0.25
p_3	0.22	0.15	0	0.15	0.28	0.11
p_4	0.37	0.20	0.15	0	0.29	0.22
p_5	0.34	0.14	0.28	0.29	0	0.39
p_6	0.23	0.25	0.11	0.22	0.39	0

HC: MIN



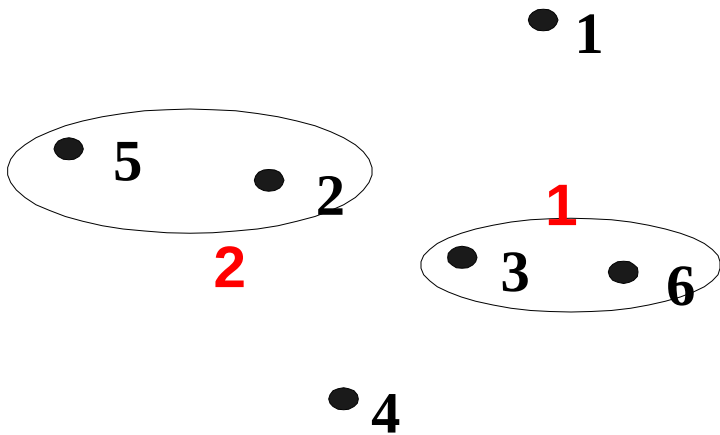
Nested Clusters

	p_1	p_2	p_3	p_4	p_5	p_6
p_1	0	0.24	0.22	0.37	0.34	0.23
p_2	0.24	0	0.15	0.20	0.14	0.25
p_3	0.22	0.15	0	0.15	0.28	0.11
p_4	0.37	0.20	0.15	0	0.29	0.22
p_5	0.34	0.14	0.28	0.29	0	0.39
p_6	0.23	0.25	0.11	0.22	0.39	0



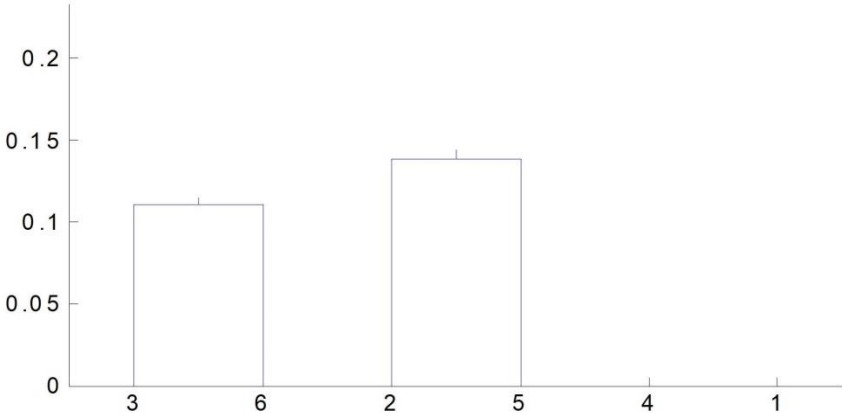
Dendrogram

HC: MIN



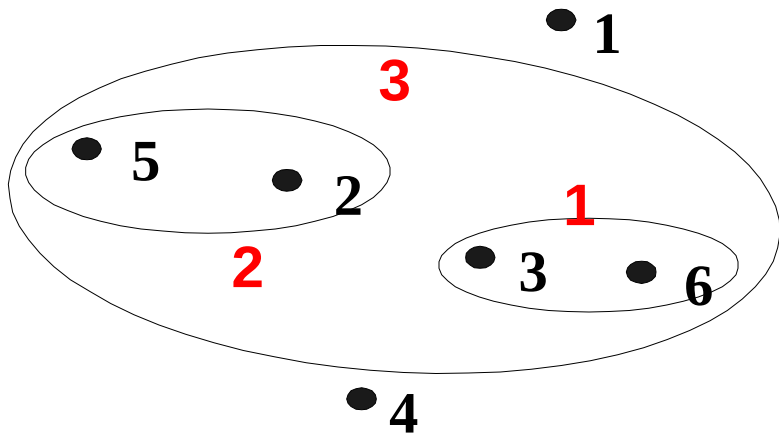
Nested Clusters

	p_1	p_2	$\{p_3, p_6\}$	p_4	p_5
p_1	0	0.24	0.22	0.37	0.34
p_2	0.24	0	0.15	0.20	0.14
$\{p_3, p_6\}$	0.22	0.15	0	0.15	0.28
p_4	0.37	0.20	0.15	0	0.29
p_5	0.34	0.14	0.28	0.29	0



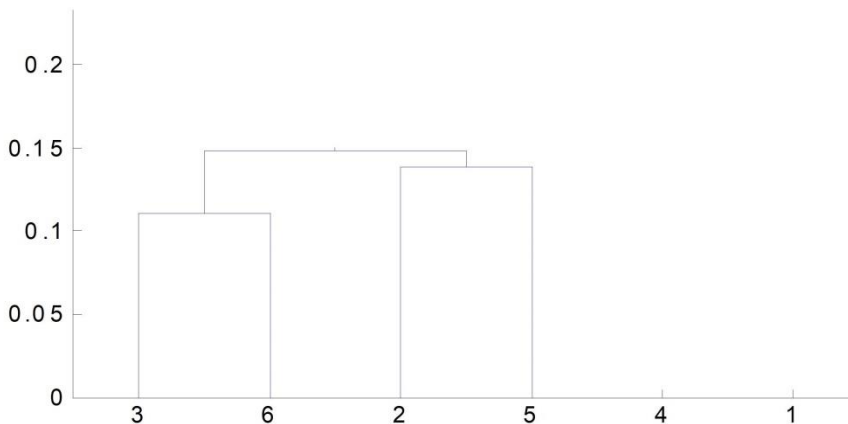
Dendrogram

HC: MIN



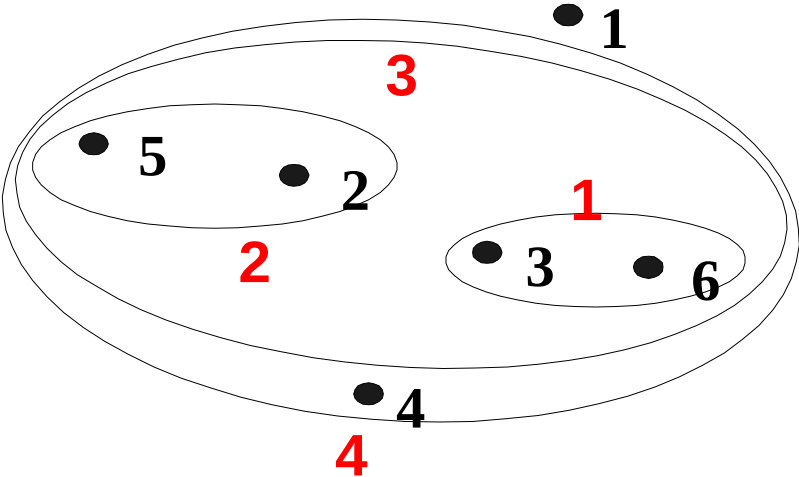
Nested Clusters

	p_1	$\{p_2, p_5\}$	$\{p_3, p_6\}$	p_4
p_1	0	0.24	0.22	0.37
$\{p_2, p_5\}$	0.24	0	0.15	0.20
$\{p_3, p_6\}$	0.22	0.15	0	0.15
p_4	0.37	0.20	0.15	0



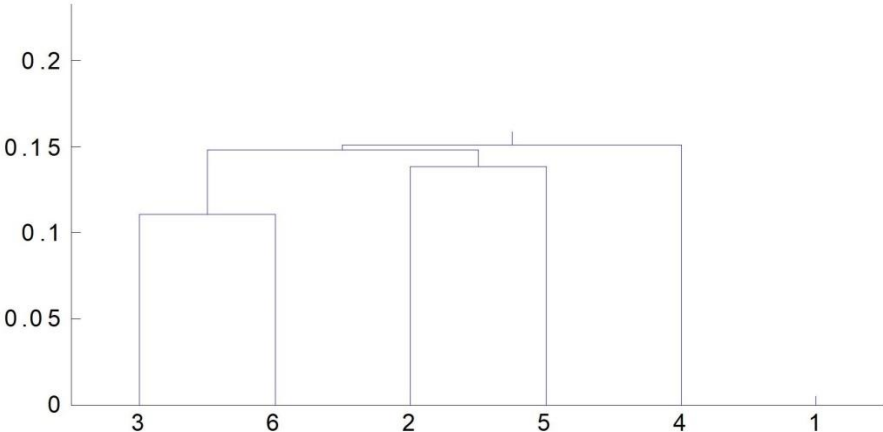
Dendrogram

HC: MIN



Nested Clusters

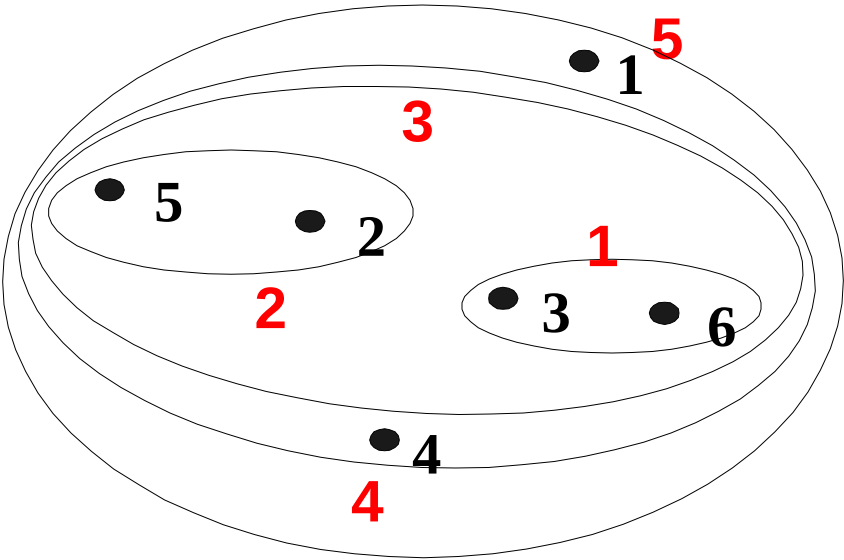
	p_1	$\{p_2, p_3, p_5, p_6\}$	p_4
p_1	0	0.22	0.37
$\{p_2, p_3, p_5, p_6\}$	0.22	0	0.15
p_4	0.37	0.15	0



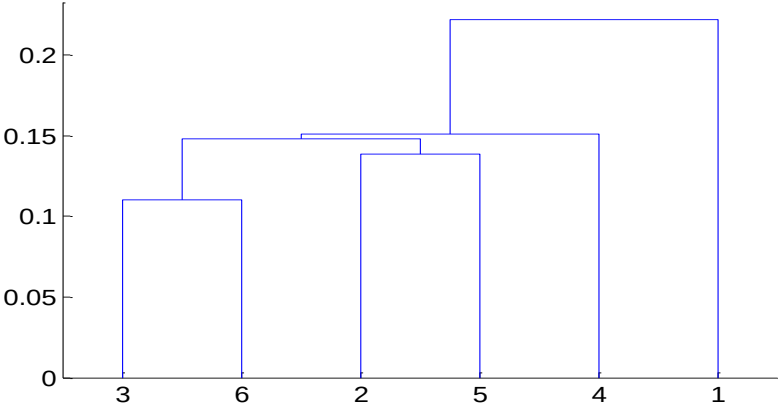
Dendrogram

HC: MIN

	p_1	$\{p_2, p_3, p_4, p_5, p_6\}$
p_1	0	0.22
$\{p_2, p_3, p_4, p_5, p_6\}$	0.22	0

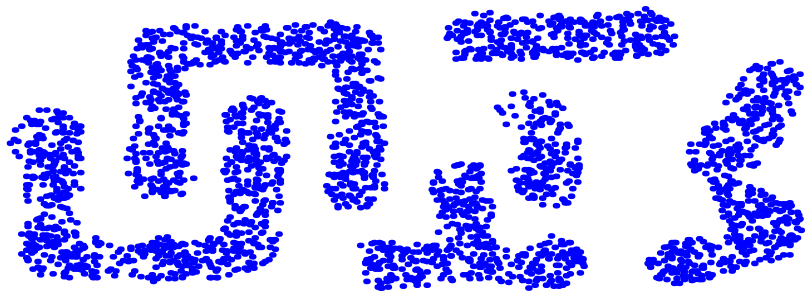


Nested Clusters

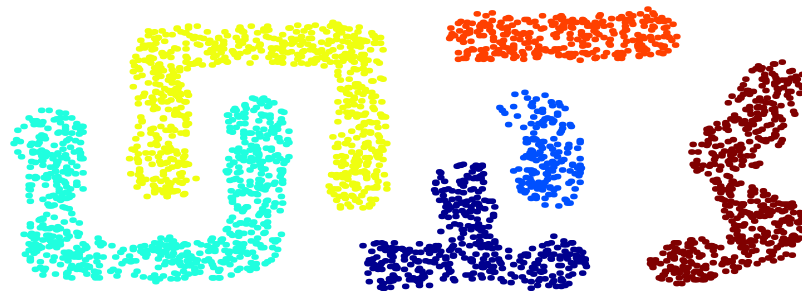


Dendrogram

STRENGTH OF MIN



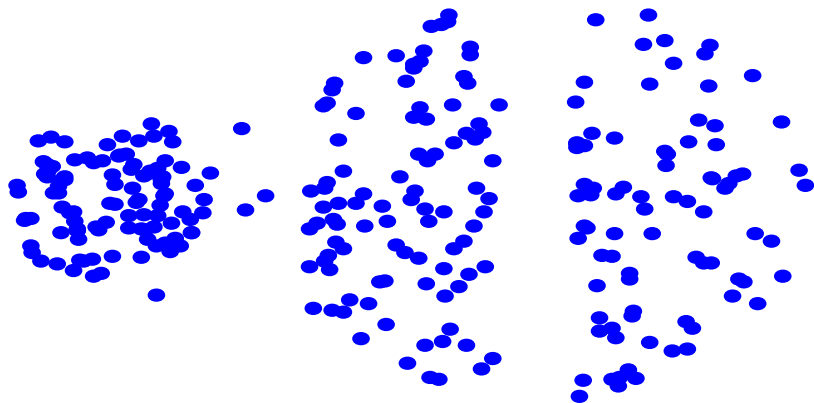
Original Points



Six Clusters

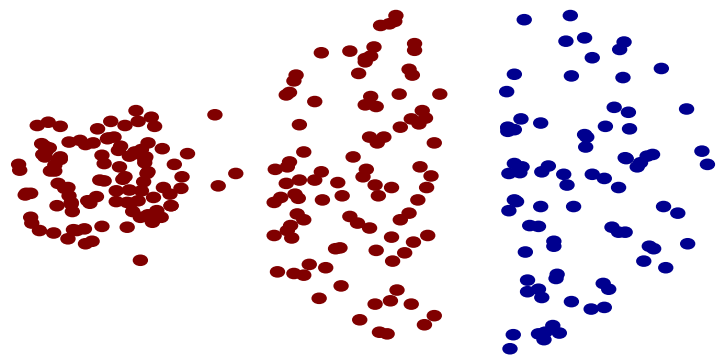
- Can handle non-elliptical shapes

LIMITATIONS OF MIN

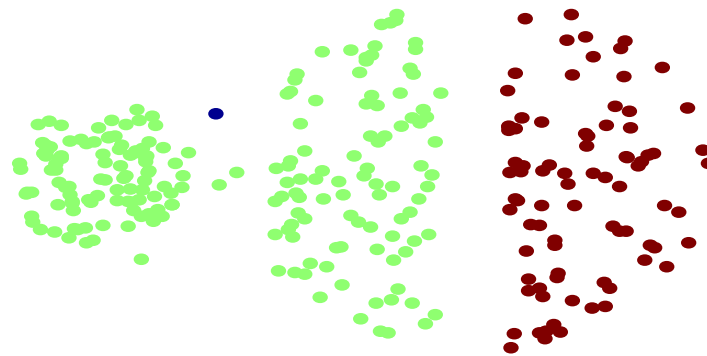


Original Points

- Sensitive to noise



Two Clusters

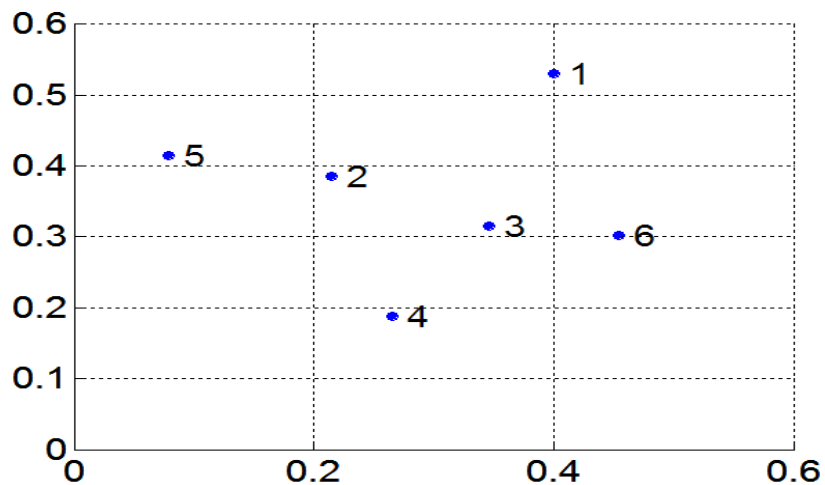


Three Clusters

MAX OR COMPLETE LINKAGE

Proximity of two clusters is based on the two most distant points in the different clusters

- Determined by all pairs of points in the two clusters

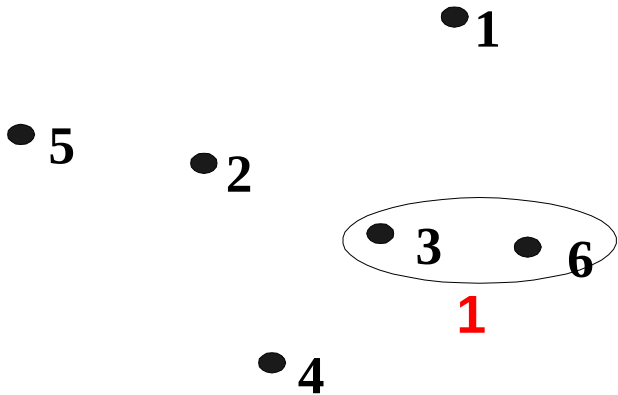


Distance Matrix:

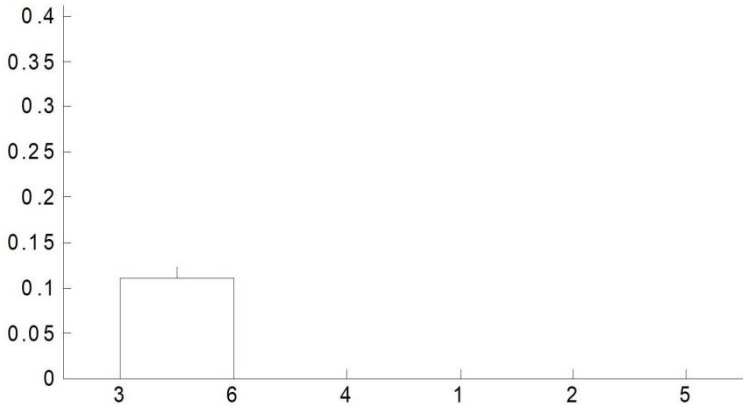
	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

HIERARCHICAL CLUSTERING: MAX

	p1	p2	p3	p4	p5	p6
p1	0.00	0.24	0.22	0.37	0.34	0.23
p2	0.24	0.00	0.15	0.20	0.14	0.25
p3	0.22	0.15	0.00	0.15	0.28	0.11
p4	0.37	0.20	0.15	0.00	0.29	0.22
p5	0.34	0.14	0.28	0.29	0.00	0.39
p6	0.23	0.25	0.11	0.22	0.39	0.00

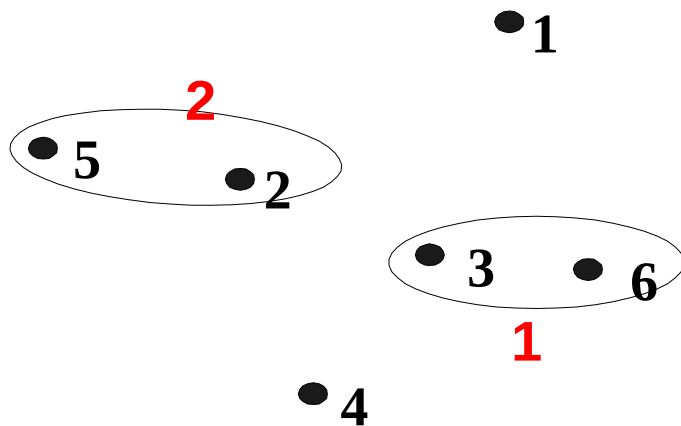


Nested Clusters



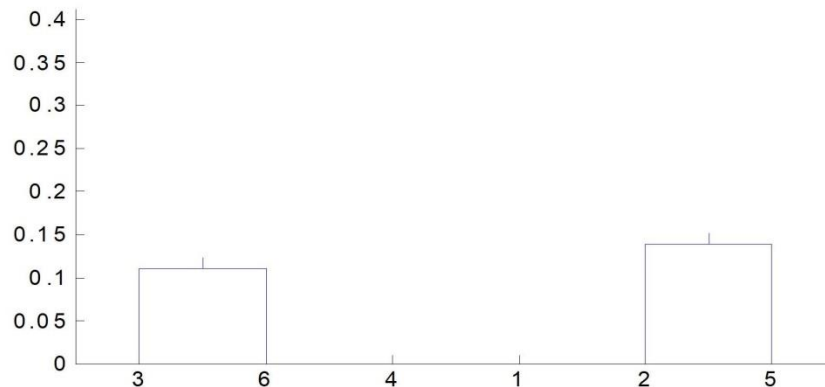
Dendrogram

HC: MAX



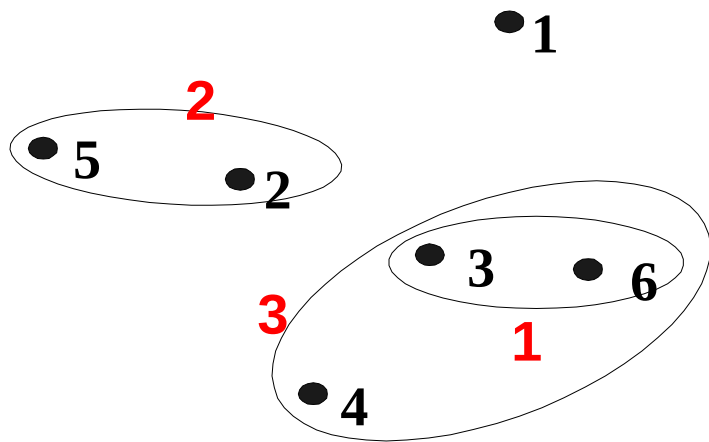
Nested Clusters

	p_1	p_2	$\{p_3, p_6\}$	p_4	p_5
p_1	0	0.24	0.23	0.37	0.34
p_2	0.24	0	0.25	0.20	0.14
$\{p_3, p_6\}$	0.23	0.25	0	0.22	0.39
p_4	0.37	0.20	0.22	0	0.29
p_5	0.34	0.14	0.39	0.29	0



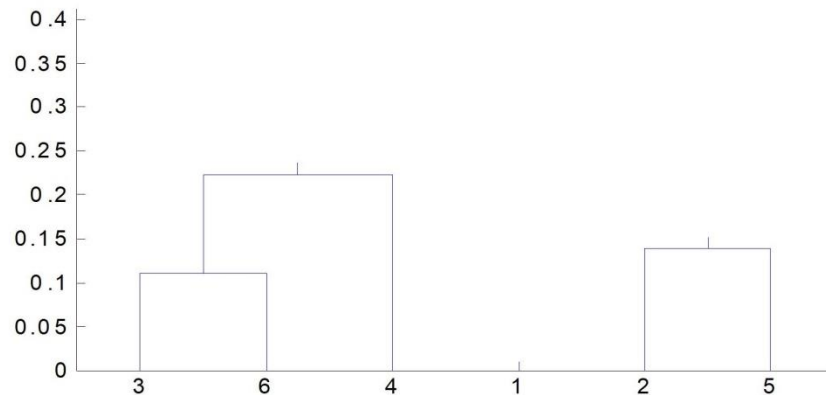
Dendrogram

HC: MAX



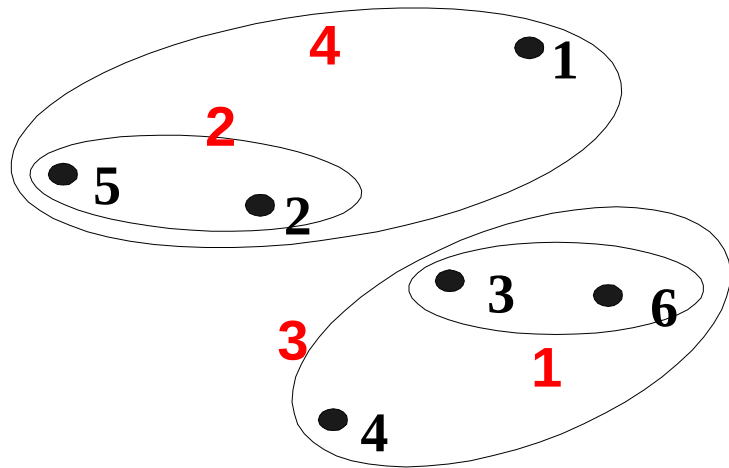
Nested Clusters

	p_1	$\{p_2, p_5\}$	$\{p_3, p_6\}$	p_4
p_1	0	0.34	0.23	0.37
$\{p_2, p_5\}$	0.34	0	0.39	0.29
$\{p_3, p_6\}$	0.23	0.39	0	0.22
p_4	0.37	0.29	0.22	0



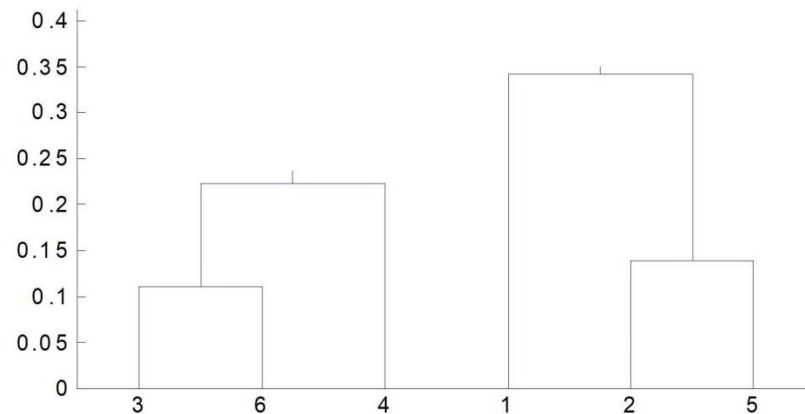
Dendrogram

HC: MAX



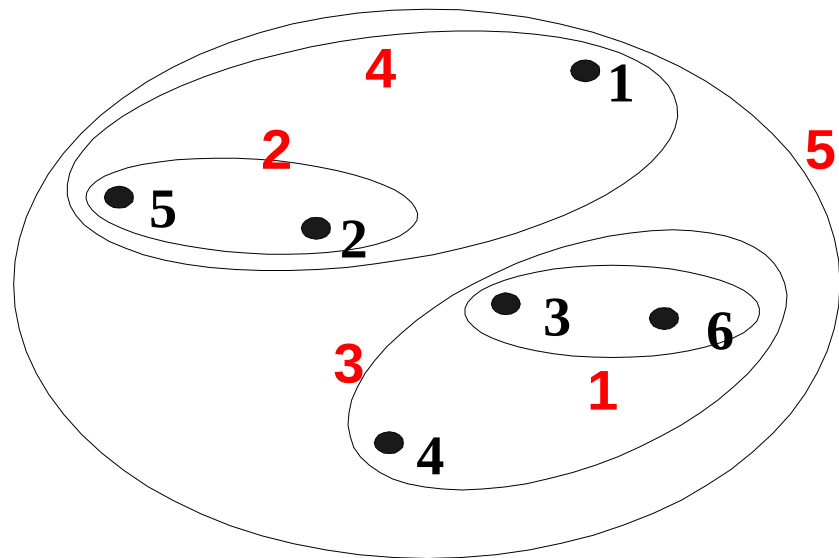
Nested Clusters

	p_1	$\{p_2, p_5\}$	$\{p_3, p_4, p_6\}$
p_1	0	0.34	0.37
$\{p_2, p_5\}$	0.34	0	0.39
$\{p_3, p_4, p_6\}$	0.37	0.39	0



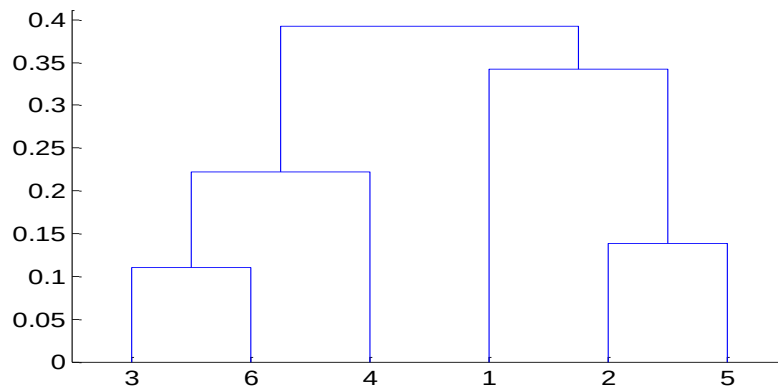
Dendrogram

HC: MAX



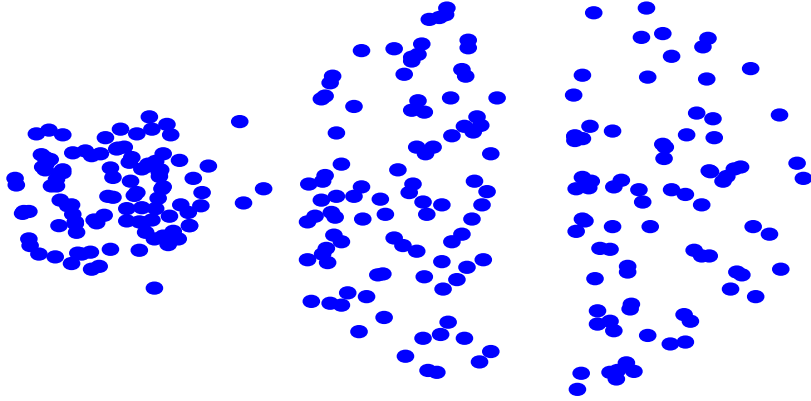
Nested Clusters

	$\{p_1, p_2, p_5\}$	$\{p_3, p_4, p_6\}$
$\{p_1, p_2, p_5\}$	0	0.39
$\{p_3, p_4, p_6\}$	0.39	0

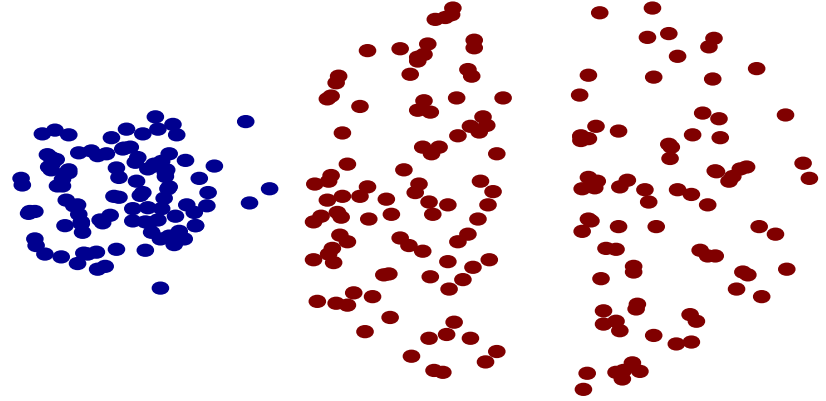


Dendrogram

STRENGTH OF MAX



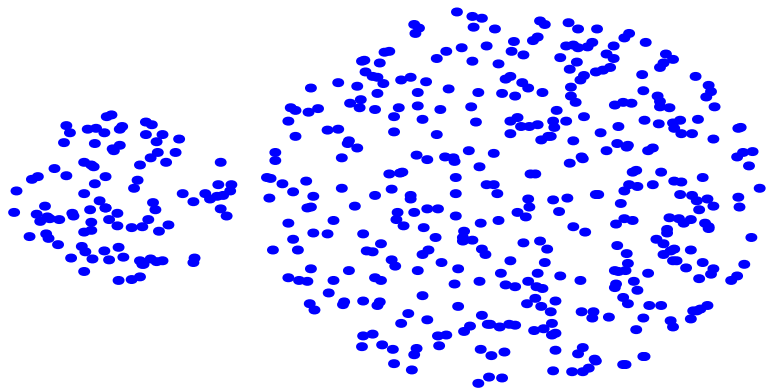
Original Points



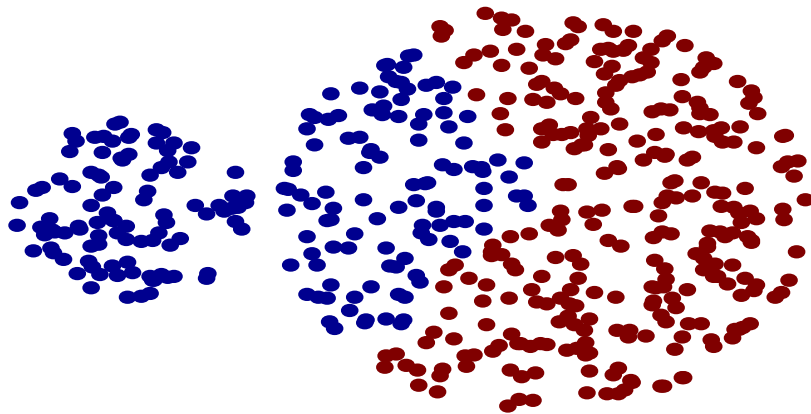
Two Clusters

- Less susceptible to noise

LIMITATIONS OF MAX



Original Points



Two Clusters

- Tends to break large clusters
- Biased towards globular clusters

HC: PROBLEMS AND LIMITATIONS

- Once a decision is made to combine two clusters, it cannot be undone
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise
 - Difficulty in handling clusters of different sizes and non-globular shapes
 - Breaking large clusters

HC EXERCISE

Exercise: Hierarchical Clustering with Single vs Complete Linkage

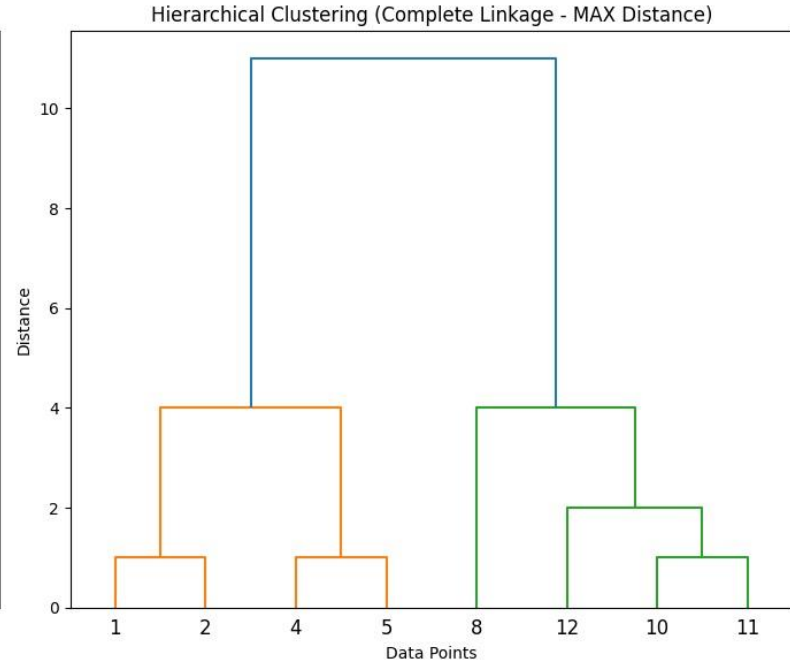
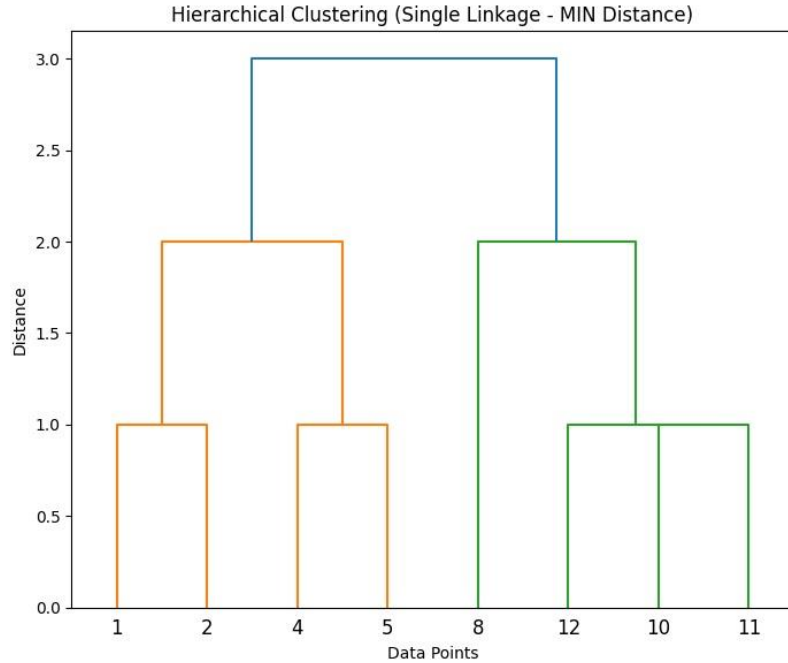
The following 1-D data points are given:

1, 2, 4, 5, 8, 10, 11, 12

Apply Agglomerative Hierarchical Clustering using two different linkage strategies:

- Single Linkage (Minimum Distance)
- Complete Linkage (Maximum Distance)

DENDROGRAMS USING SINGLE AND COMPLETE LINKAGE



RECOMMENDER SYSTEMS

RECOMMENDER SYSTEMS

- When Netflix suggests movies for you or Amazon shows 'Customers who bought this also bought,' that's a recommender system at work.
- Recommender systems types
 - Collaborative filtering
 - Content-based
 - Knowledge-based
 - Hybrid

Reference: Recommender Systems, An Introduction: Dietmar Jannach, Markus Zanker, Alexander Felfernig and Gerhard Friedrich

COLLABORATIVE FILTERING (CF)

- **Core Idea:** If two users liked similar items in the past, they are likely to have similar tastes in the future.
- **Recommendation Logic:** If user A and user B have a large overlap in purchase/usage history, and A likes something new, the system recommends that item to B.
- **Why the name “Collaborative” Filtering:** Users indirectly help each other find items — their collective behavior guides recommendations.
- **No Need for Item Details:** CF does not require knowledge of item attributes (genre, author, features, etc.) — it relies purely on user behavior (ratings, clicks, purchases).

COMMON CHALLENGES

- **Finding similar users:** How do we determine whose tastes are alike?
- **Similarity Measurement:** Which metric should be used (e.g., cosine similarity, correlation)?
- **Cold Start Problem:**
- **New User:** No history to base recommendations on
- **New Item:** Nobody has rated it yet
- **Data Sparsity:** What if only a few ratings exist in the system?
- **Alternative Approaches:** Beyond finding similar users, models can learn patterns to predict ratings.

CF: ADVANTAGES AND LIMITATIONS

- Advantages

- ☐ No need to manually label or describe items.
- ☐ Works well when large amounts of user interaction data are available.

- Limitation

- ☐ Ignores *content* of items — sometimes using features like genre or author can improve recommendation quality (→ leads to Hybrid systems).

CONTENT BASED (CB)

- Two purposes of recommender systems:
 - ❑ To influence user choices (e.g., suggest books or movies to explore).
 - ❑ To reduce information overload by selecting relevant items from large collections.
- Connection to Information Retrieval:
 - ❑ Similar to filtering relevant vs. irrelevant documents.
 - ❑ Recommendations are based on item content and how well it matches user interests.

CONTENT BASED (CB)

- Core Principle:

- ☐ Each item has descriptive features (e.g., genre, topic, author).
- ☐ A user profile represents what the user likes.
- ☐ The system recommends items that have similar attributes to the items the user has liked before.

- Building User Profiles:

- ☐ Can be learned automatically from user behavior (ratings, clicks).
- ☐ Or derived from explicit user input on preferences.

KEY ISSUES (CB)

How to learn and update user profiles over time?

How to measure similarity between items and user interests?

How to automatically extract item features, to avoid manual tagging?

CB: ADVANTAGES AND LIMITATIONS

- Advantages:

- ☐ Works even with few users (no need for large user groups).
- ☐ New items can be recommended immediately if their attributes are known.

- Limitations:

- ☐ Hard to represent subjective qualities (e.g., elegance, emotional impact).
- ☐ Such attributes often require manual input, which can be costly and error-prone.

KNOWLEDGE BASED (KB)

- When it is needed:
 - ❑ Used in domains where users do not buy frequently (e.g., digital cameras, cars, appliances).
 - ❑ No purchase history is available → collaborative filtering and content-based systems are not effective.
- Core Idea:
 - ❑ Recommendations are based on explicit knowledge about:
 - Item features (e.g., price, resolution, weight).
 - User requirements & constraints (e.g., budget limit, preferred image quality).

KNOWLEDGE BASED (KB)

- Constraint-Based Recommendation:

- ☐ The system applies rules and conditions to match items with user needs.
- ☐ Example: *Recommend high-resolution cameras if the user wants large printable photos.*

- Personalization Still Required:

- ☐ Simply filtering products by features is not enough → all users would receive the same results.
- ☐ The system must ask the user which features matter more, e.g., “Is resolution more important than weight?”

- User Interaction is Central:
 - ❑ The system often interacts with the user to clarify preferences.
 - ❑ Useful for first-time users with no behavioral history.
 - ❑ May involve:
 - Asking questions (max price, minimum resolution, etc.)
 - Conversational or guided dialogs to reduce cognitive load.

KB: CHALLENGES

- How to represent domain knowledge in the system?
- How to rank items based on user requirements and priorities?
- How to learn or build user profiles without past behavior?
- How to design interactive dialogs that help users express preferences easily?

HYBRID

- Different recommender techniques (Collaborative, Content-Based, Knowledge-Based) each have strengths and weaknesses.
- Hybrid systems combine two or more approaches to improve accuracy and reduce limitations.
- Why Hybridization Helps:
 - ❑ Helps solve the cold start problem (e.g., new users or new items) by using content features when behavioral data is missing.
 - ❑ Can provide better, more balanced recommendations.

HYBRID

- Combine Collaborative Filtering (learns from user behavior) with Content-Based methods (uses item attributes).
- Useful when both user preference data and item feature information are available.
- Key Design Questions in Hybrid Systems:
 - ☐ Which techniques should be combined?
 - ☐ Should the methods be applied in sequence, or merged into a unified model?
 - ☐ How should the outputs be weighted, and can weights adjust automatically?

ANN: ARTIFICIAL NEURAL NETWORKS

ANN

Supervised Approach

- ANN used for regression and classification
- CNN used for computer vision
- RNN used for time series analysis

THE NEURON

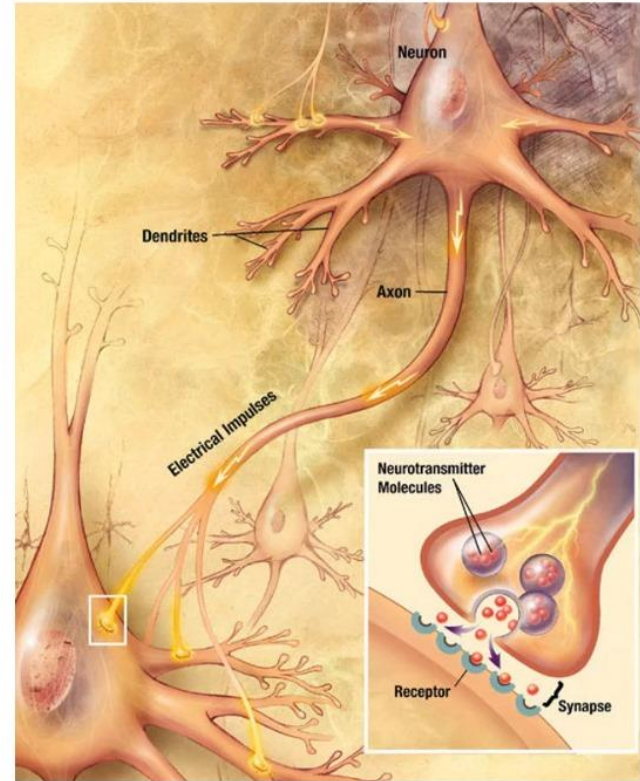
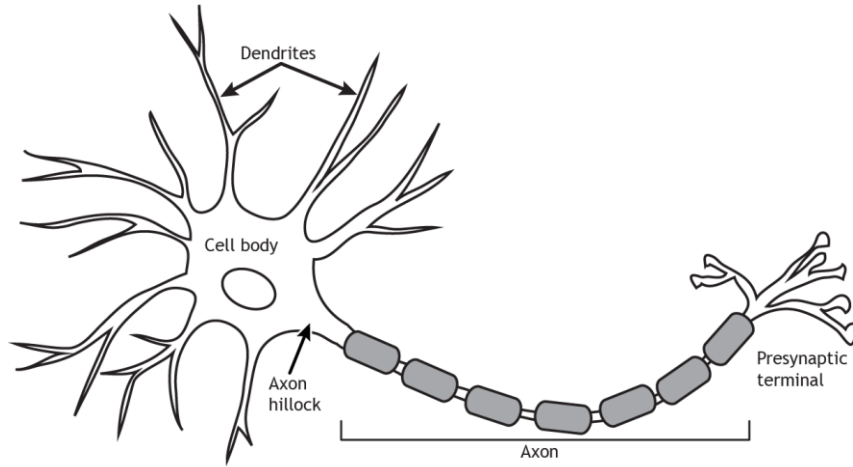
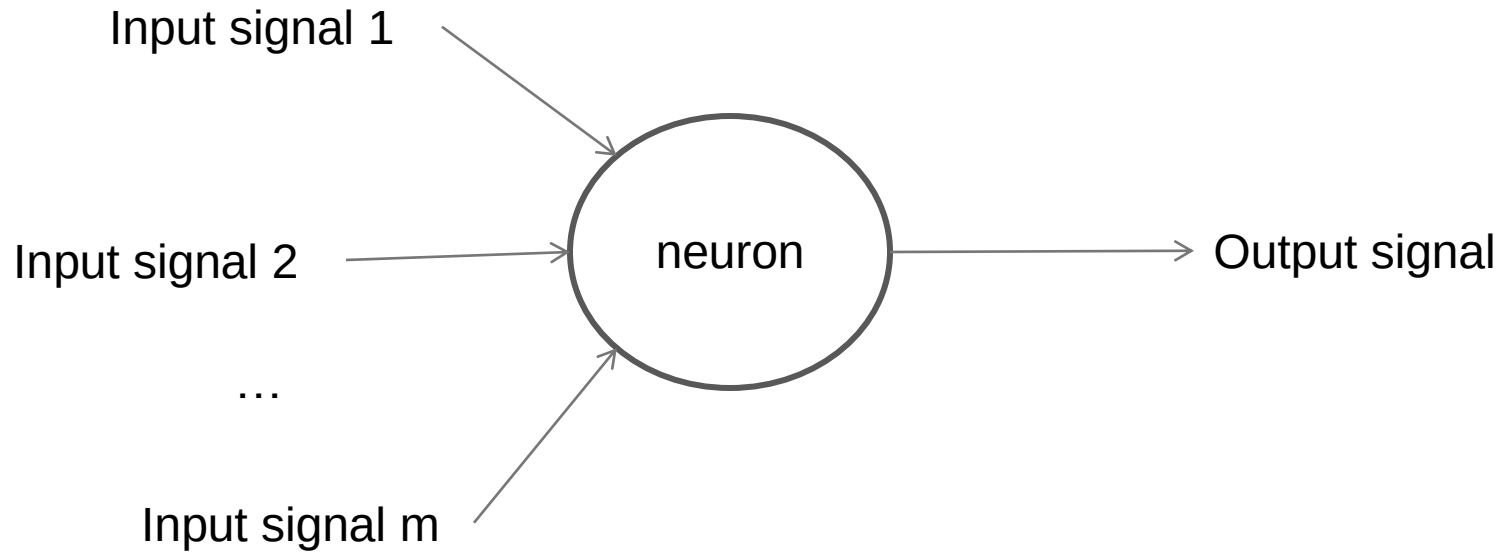
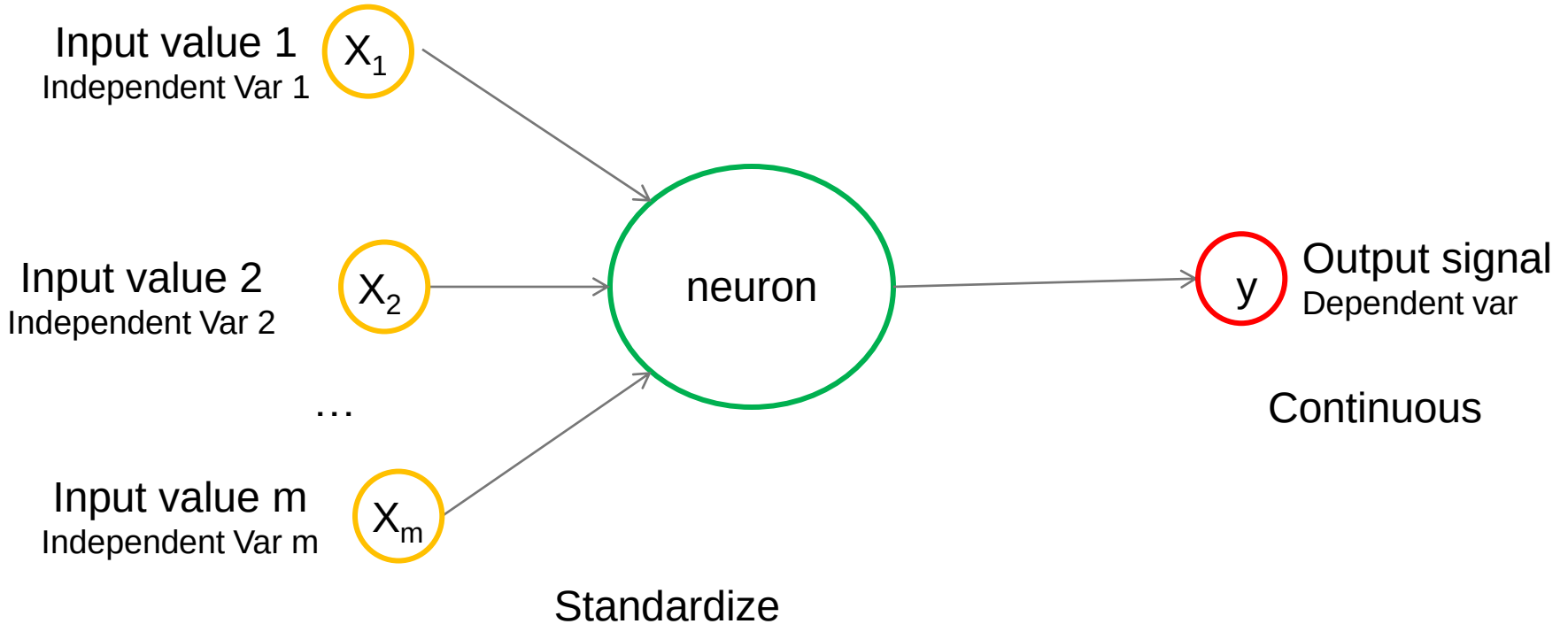


Image Source: Wikipedia

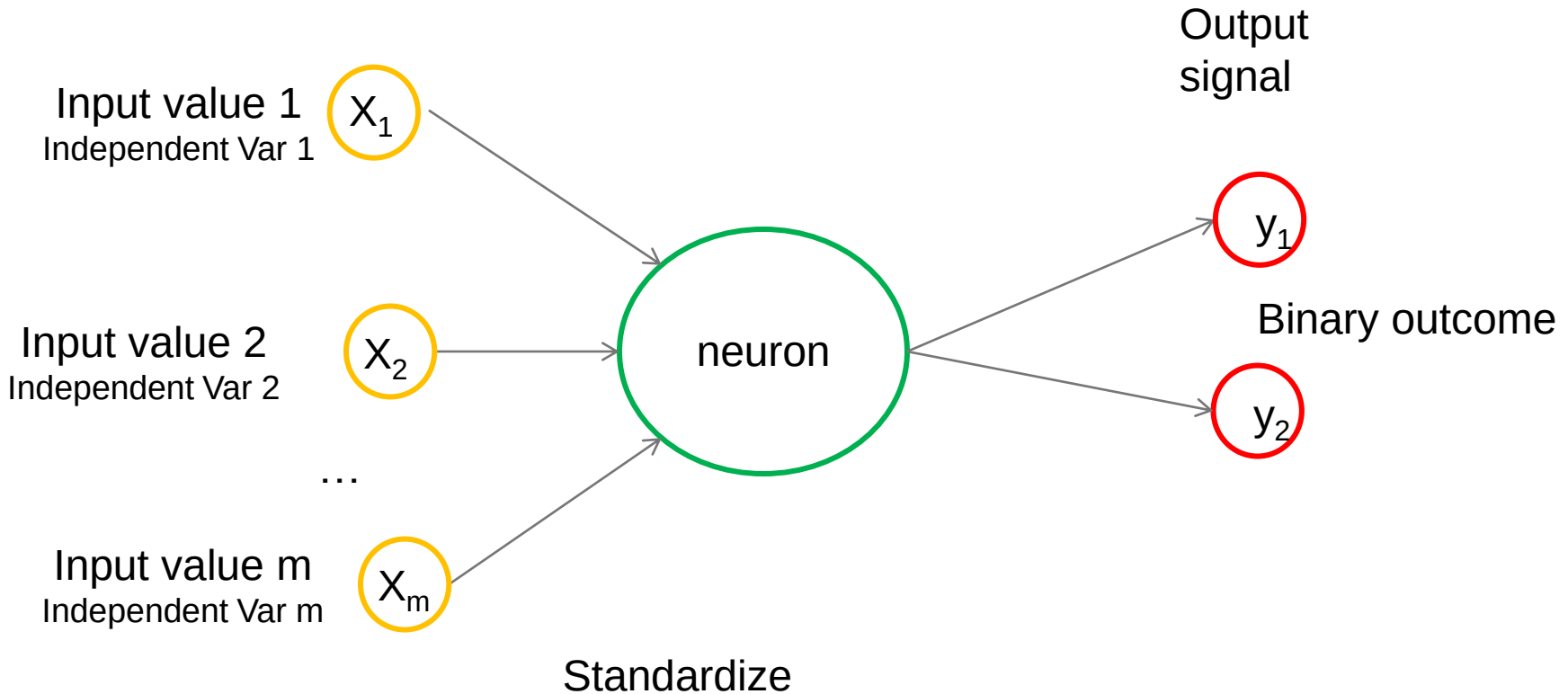
THE NEURON



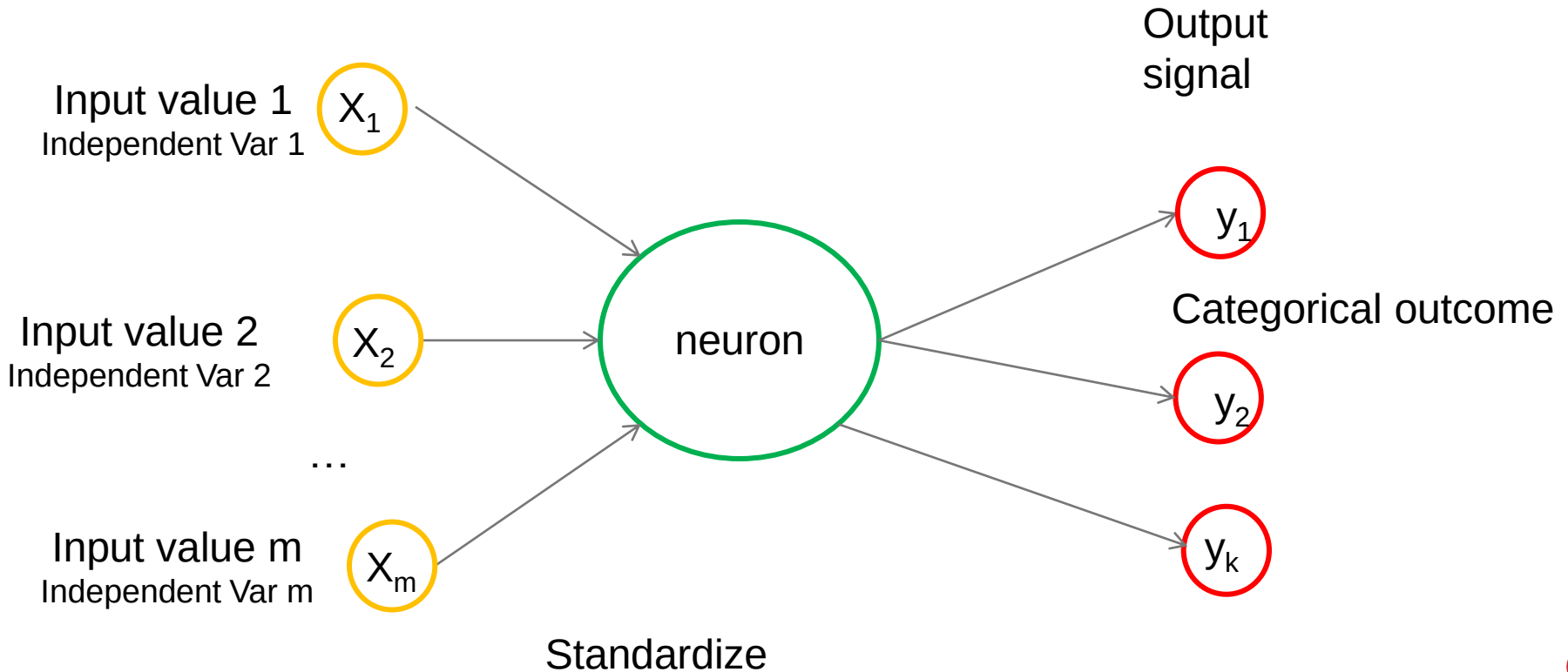
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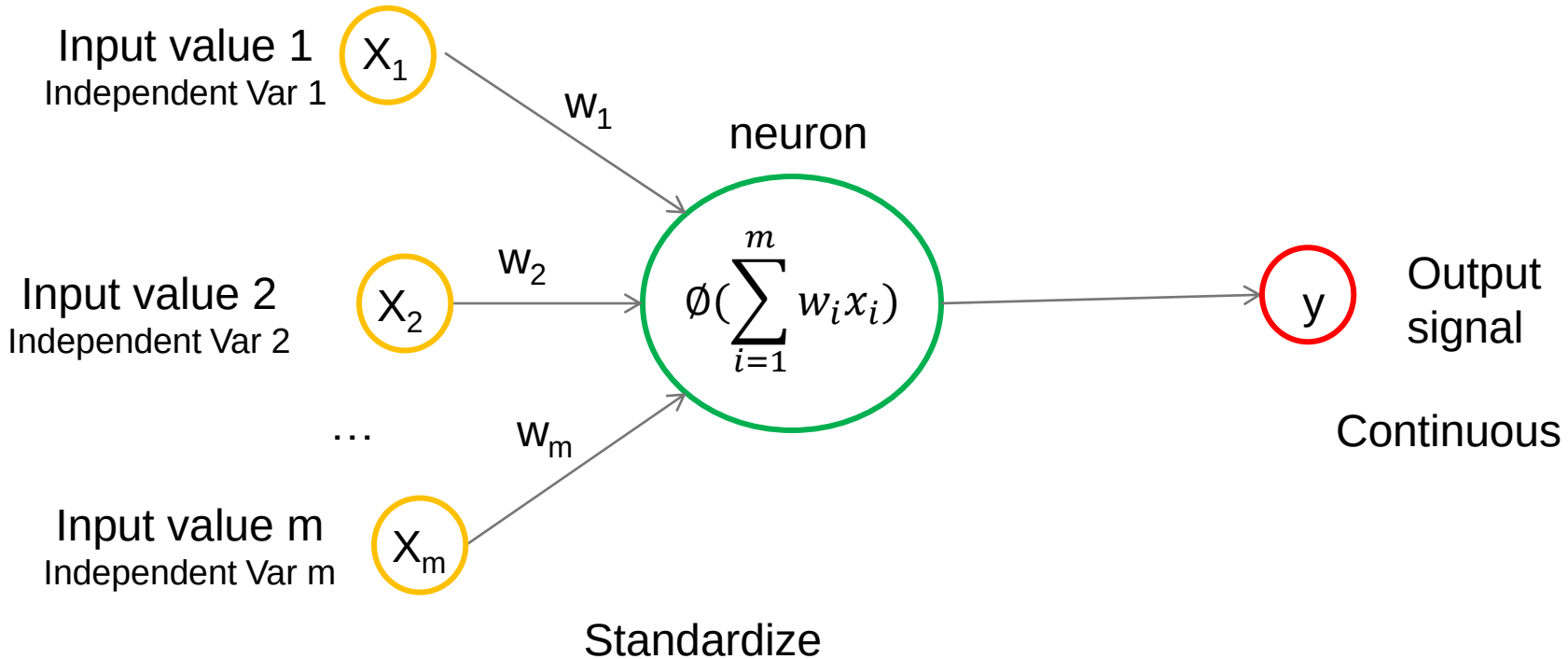
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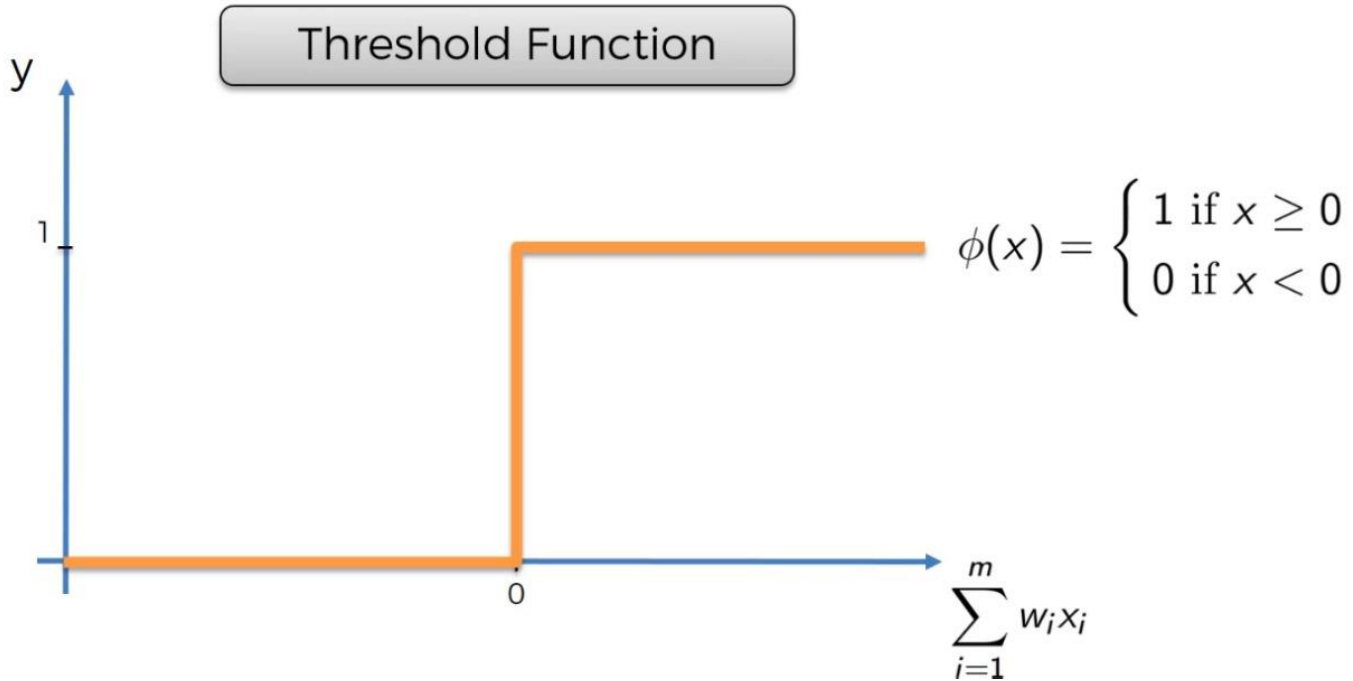
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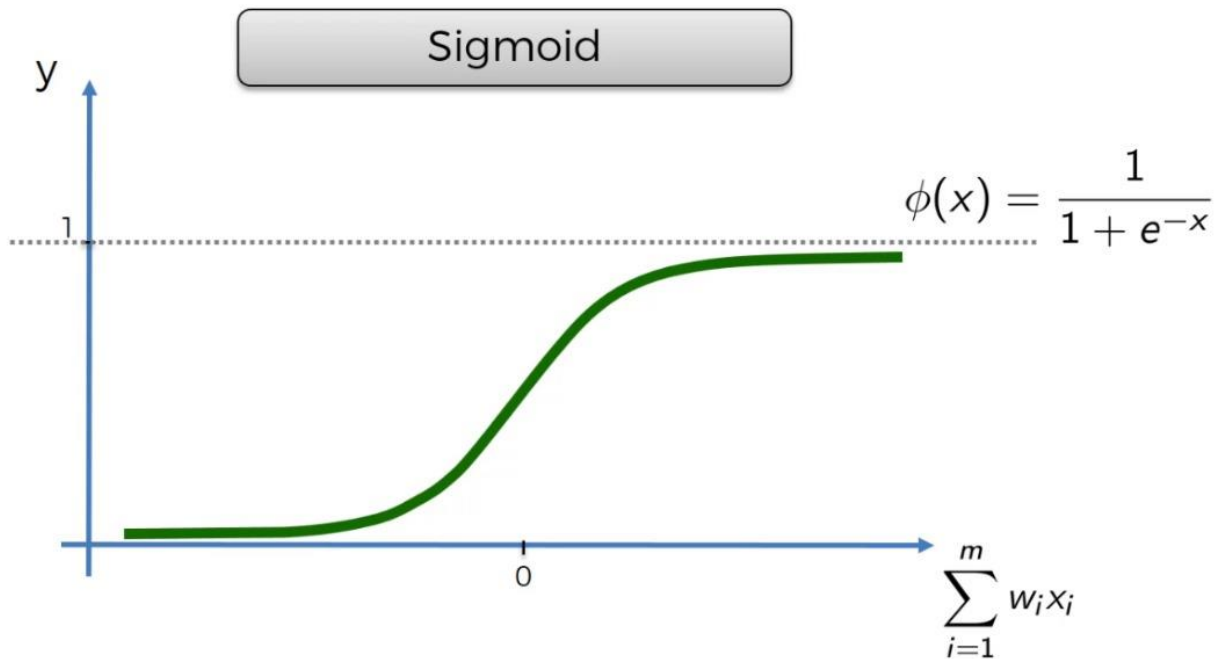
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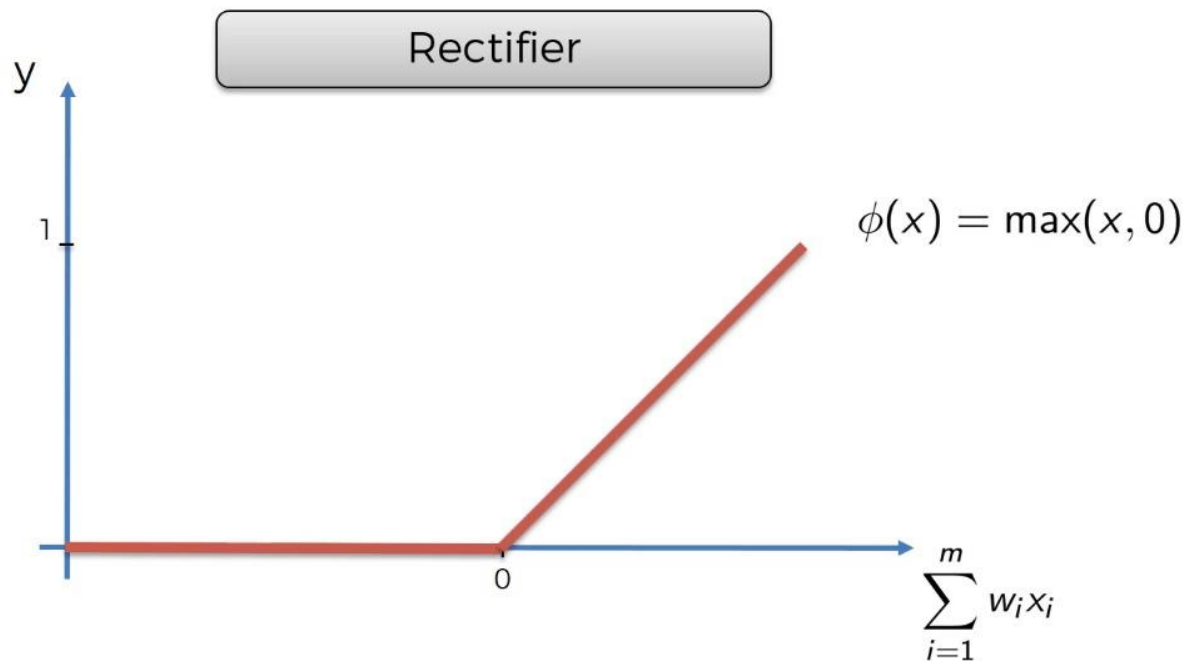
THE ACTIVATION FUNCTION



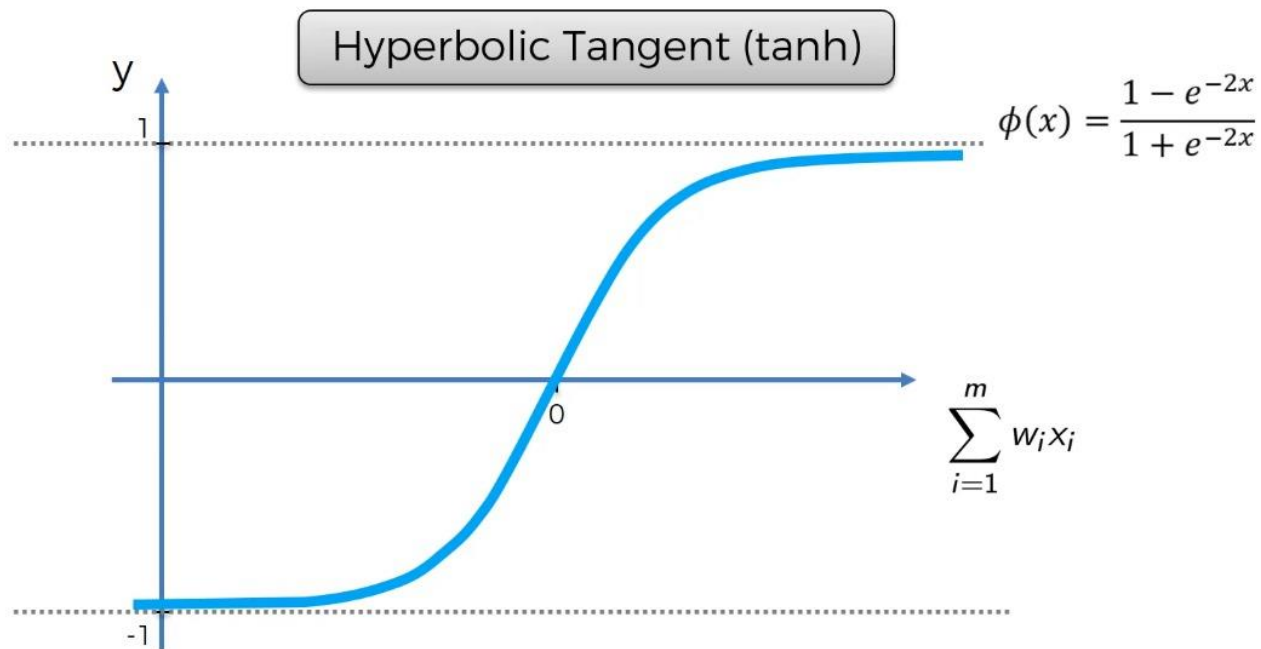
THE ACTIVATION FUNCTION



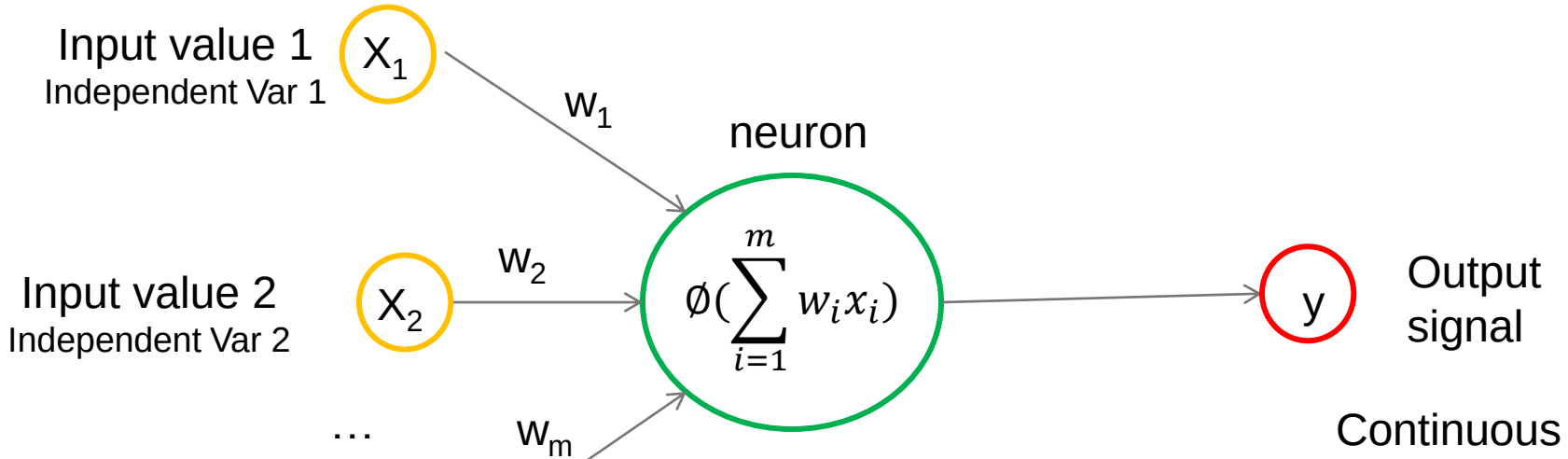
THE ACTIVATION FUNCTION



THE ACTIVATION FUNCTION

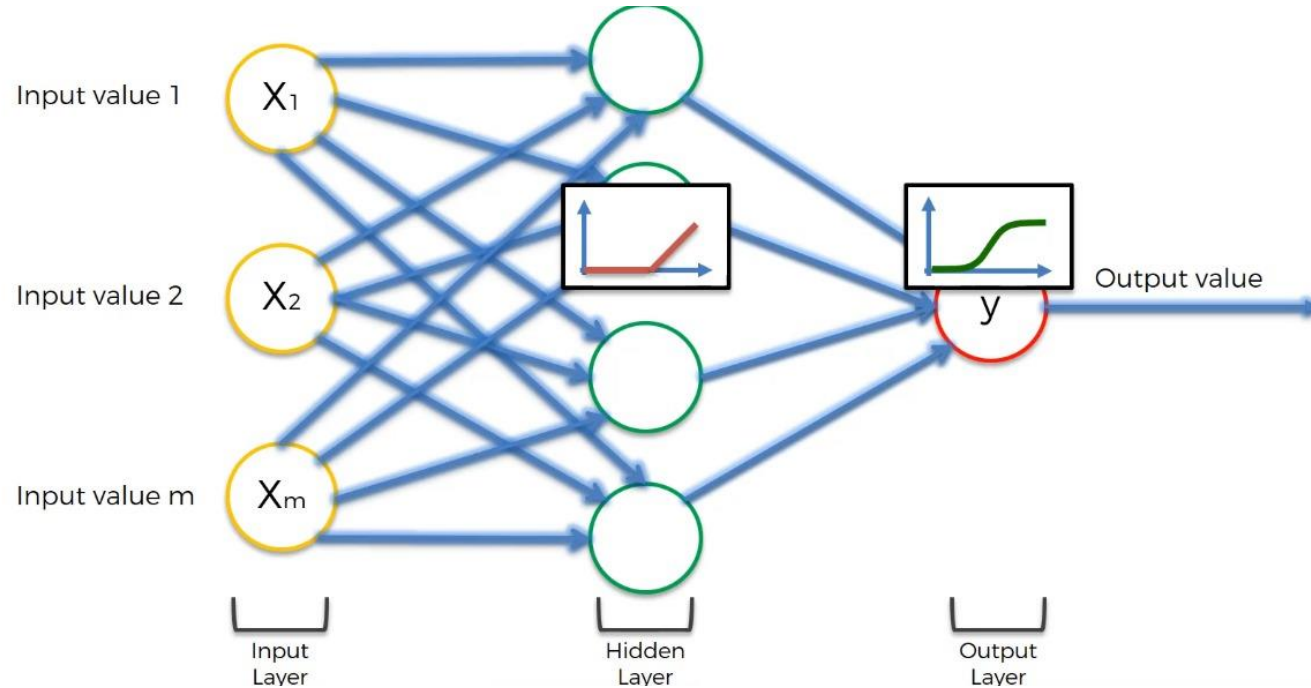


THE ACTIVATION FUNCTION

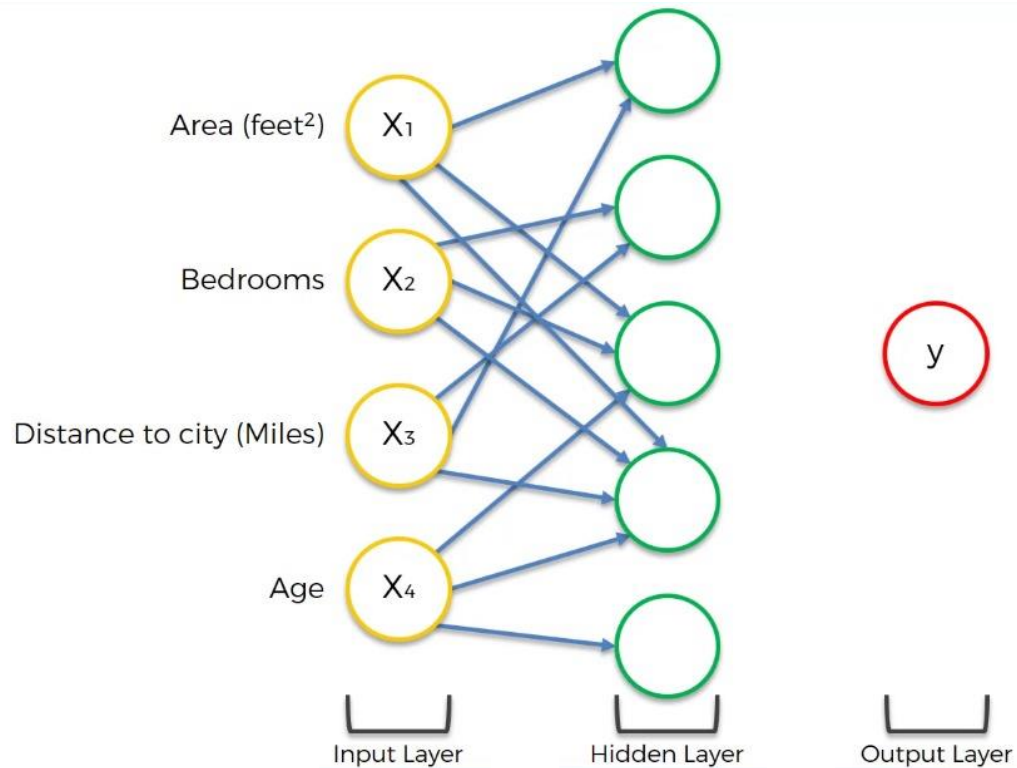


Assuming Dependent variable is binary, what activation function can be used?

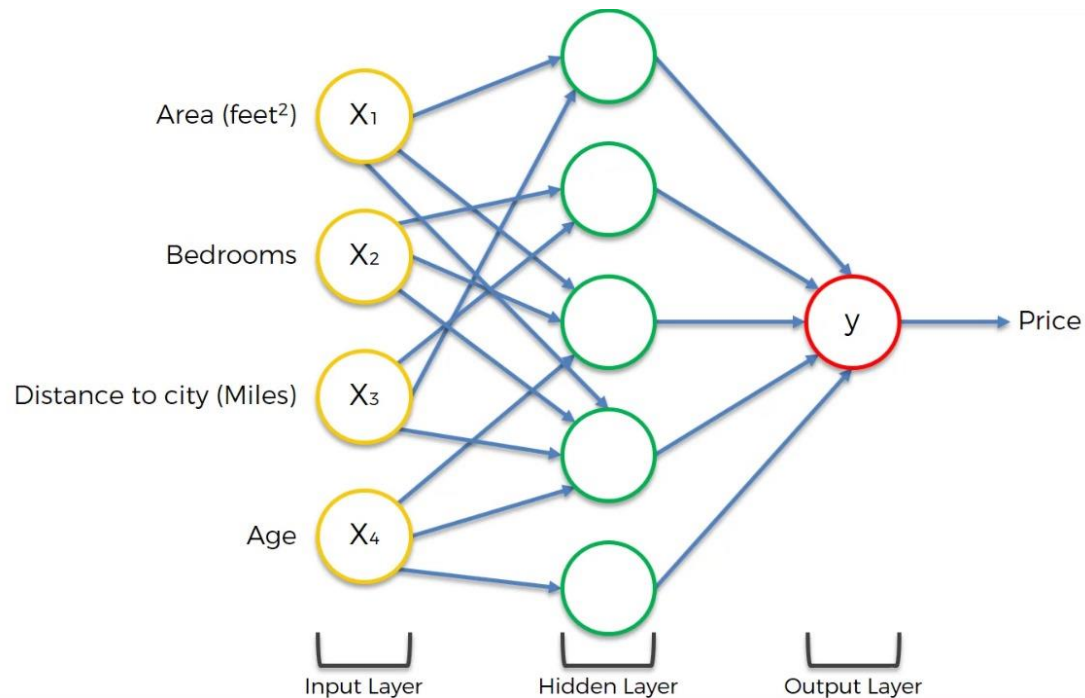
THE ACTIVATION LAYER



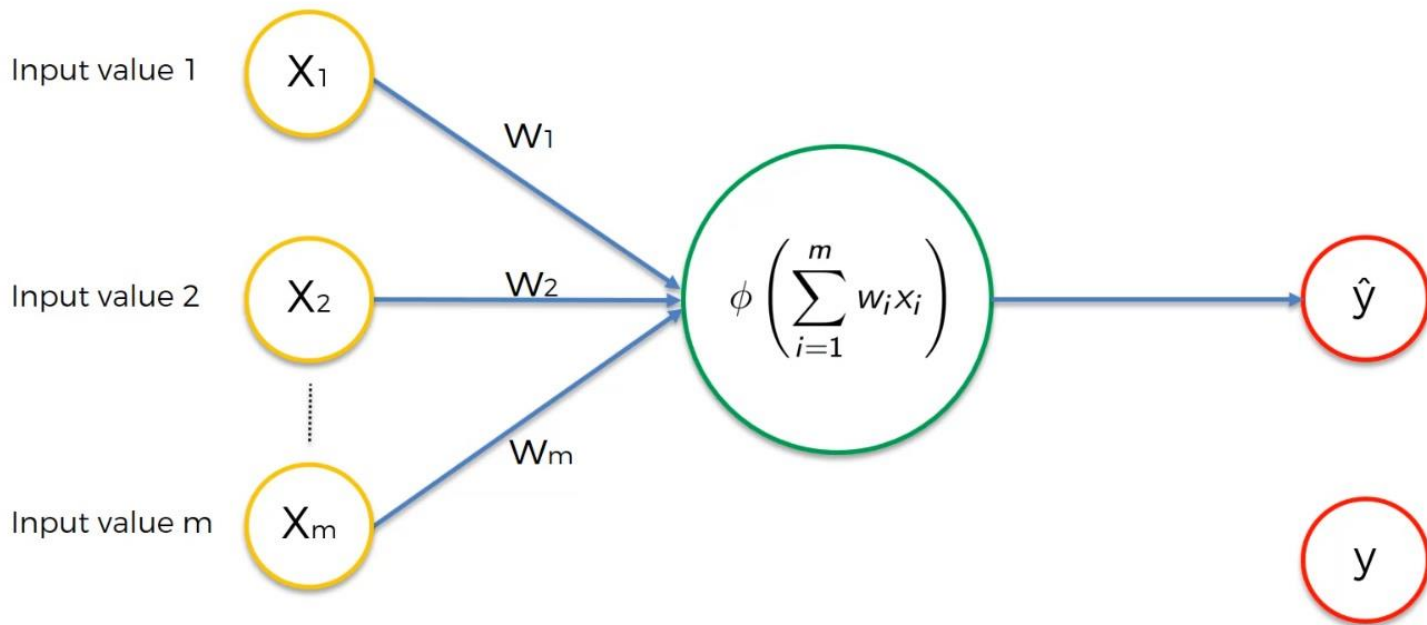
HOW NN WORKS?



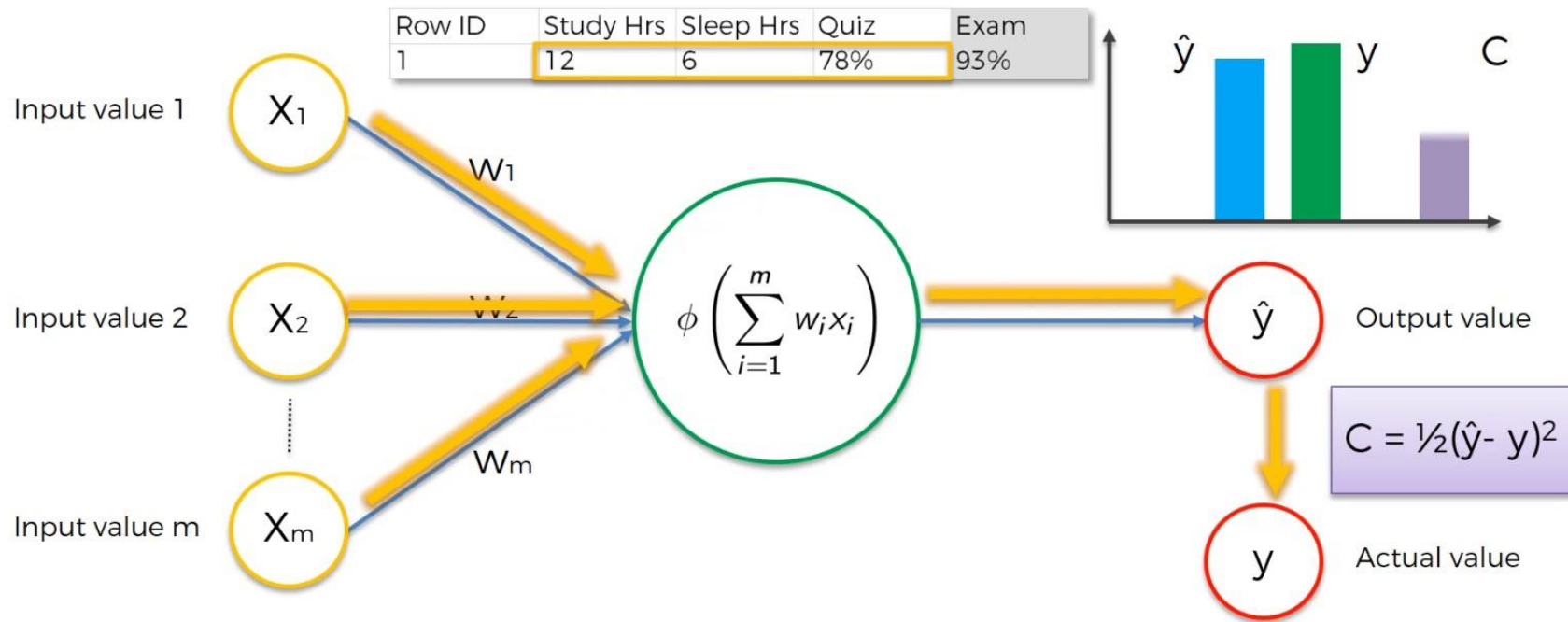
HOW NN WORKS?



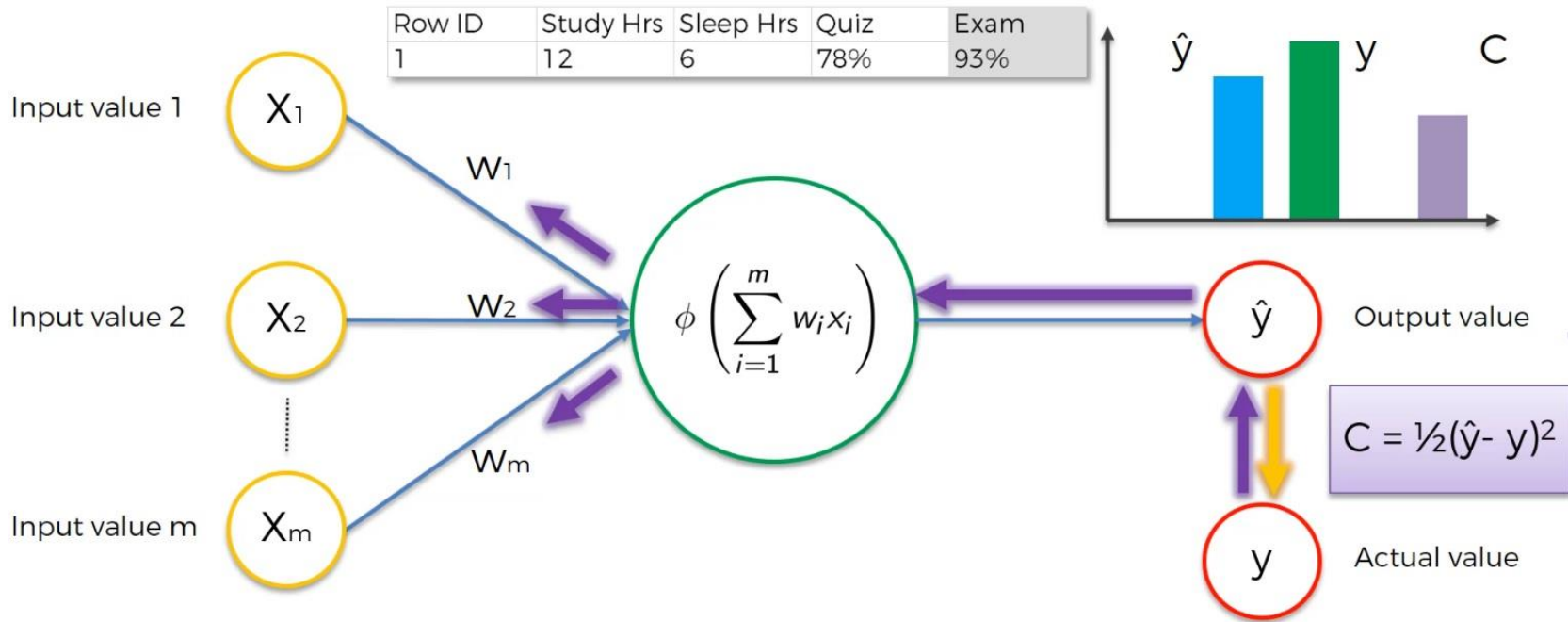
HOW NN WORKS?



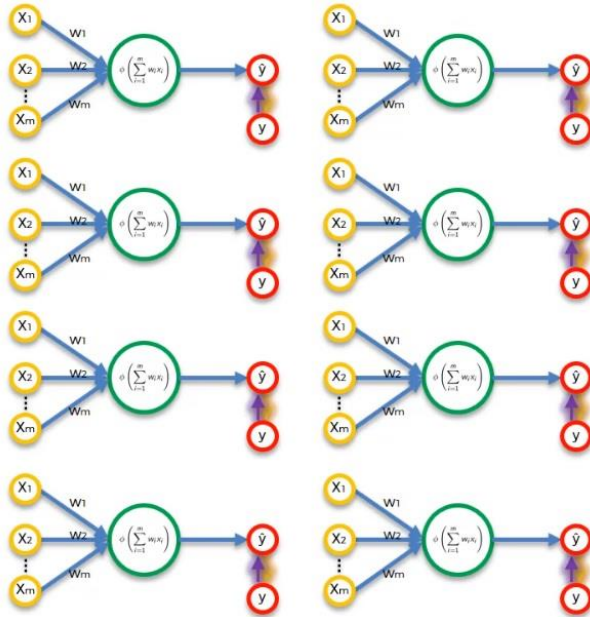
HOW NN WORKS?



HOW NN WORKS?



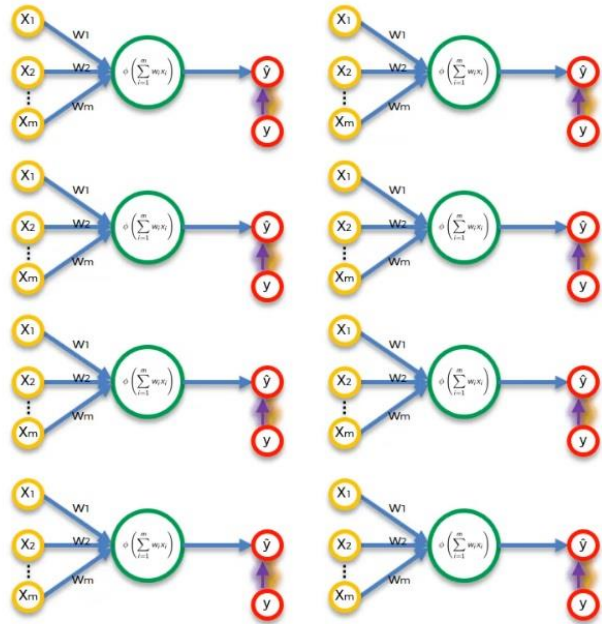
HOW NN WORKS?



Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

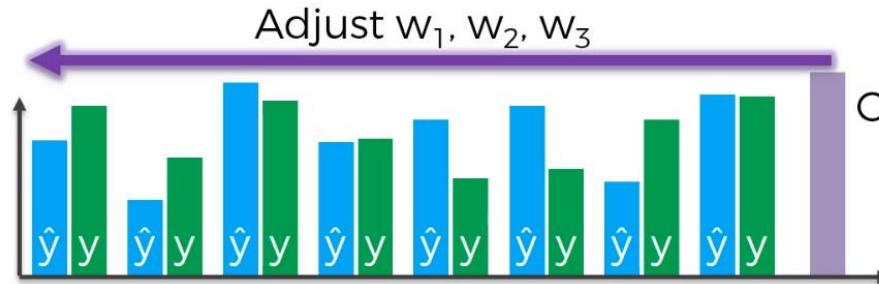


HOW NN WORKS?



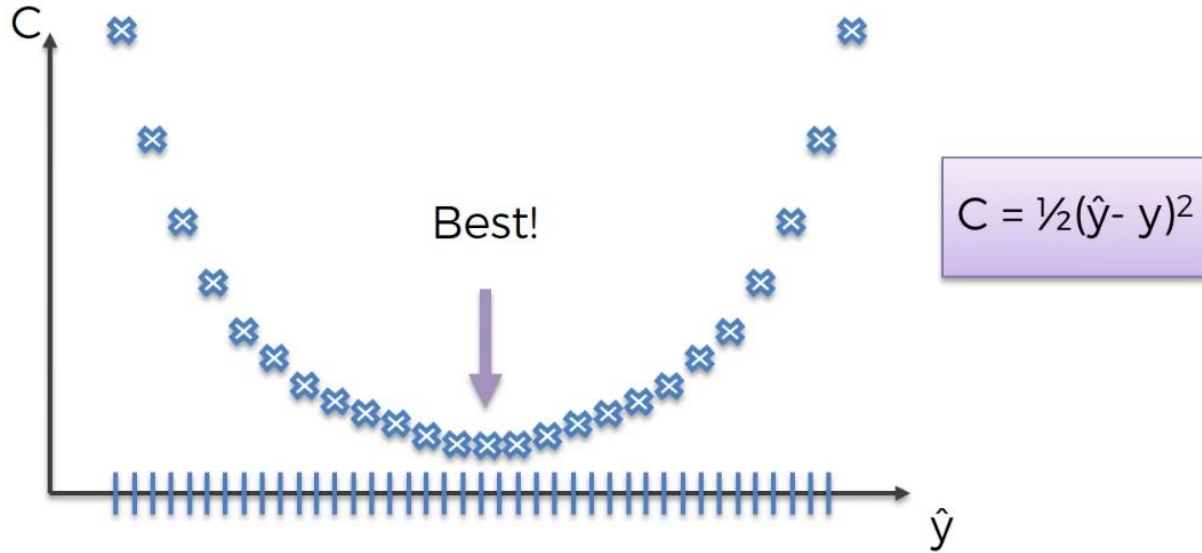
Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

$$C = \sum \frac{1}{2}(\hat{y} - y)^2$$

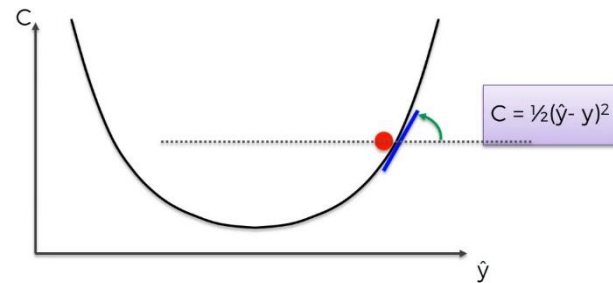
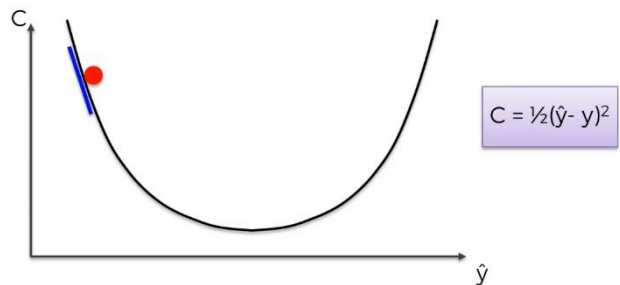


Minimize C

BRUTE FORCE



GRADIENT DESCENT



BACKPROPAGATION

We calculate how much each weight contributed to the error.

The

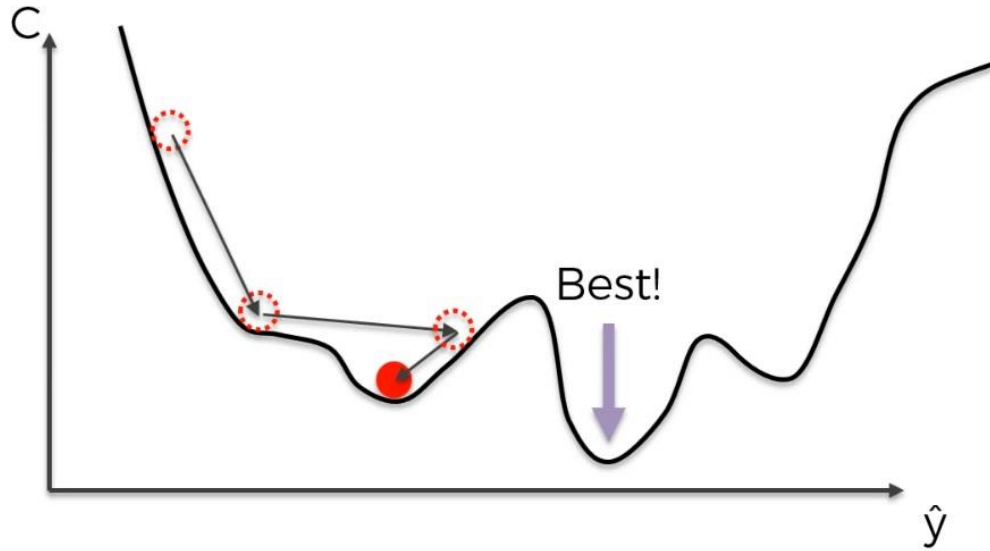
$$\text{New Weight} = \text{Old Weight} - \eta \frac{\partial \text{loss}}{\partial \text{weight}}$$

Where:

η = Learning Rate (small positive number)

Partial derivative tells how sensitive the error is w.r.t that weight.

STOCHASTIC GRADIENT DESCENT



HOW TWO ARE DIFFERENT

Upd w's ←

Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

Batch
Gradient
Descent

Upd w's ←

Upd w's ←

Upd w's ←

Upd w's ←

Upd w's ←

Upd w's ←

Upd w's ←

Upd w's ←

Row ID	Study Hrs	Sleep Hrs	Quiz	Exam
1	12	6	78%	93%
2	22	6.5	24%	68%
3	115	4	100%	95%
4	31	9	67%	75%
5	0	10	58%	51%
6	5	8	78%	60%
7	92	6	82%	89%
8	57	8	91%	97%

Stochastic
Gradient
Descent

MINI BATCH GRADIENT DESCENT

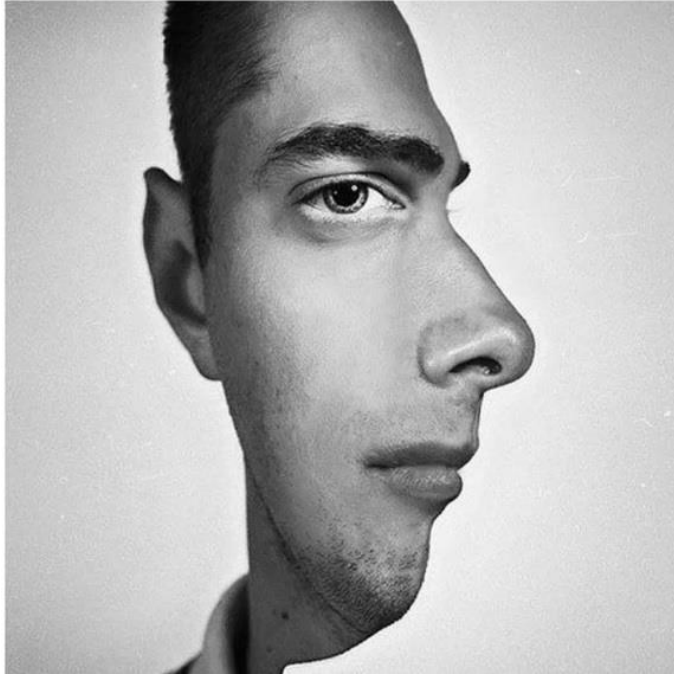
Uses small batches (e.g., 32 or 64 samples).

Most common in modern machine learning (especially deep learning).

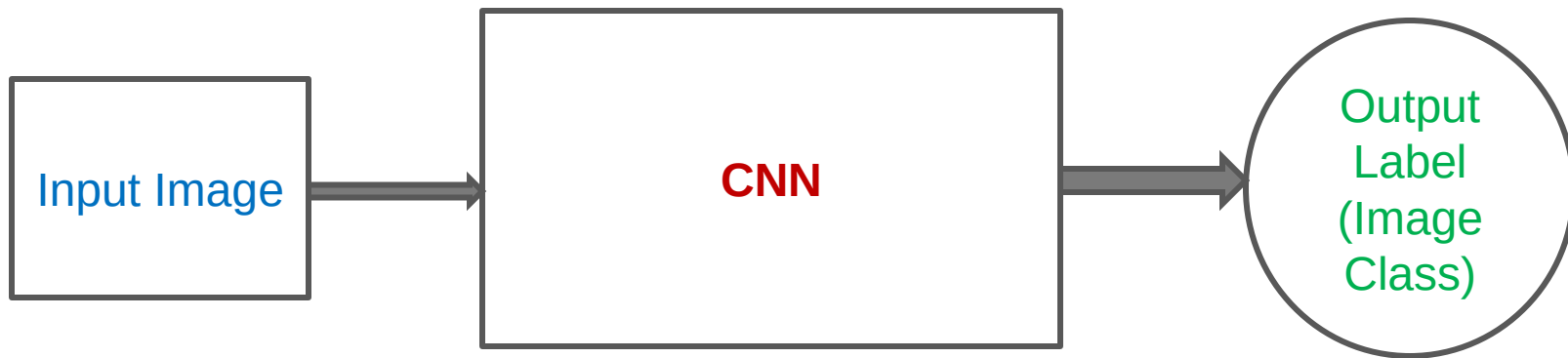
Combines speed and stability.

CNN: CONVOLUTIONAL NEURAL NETWORKS

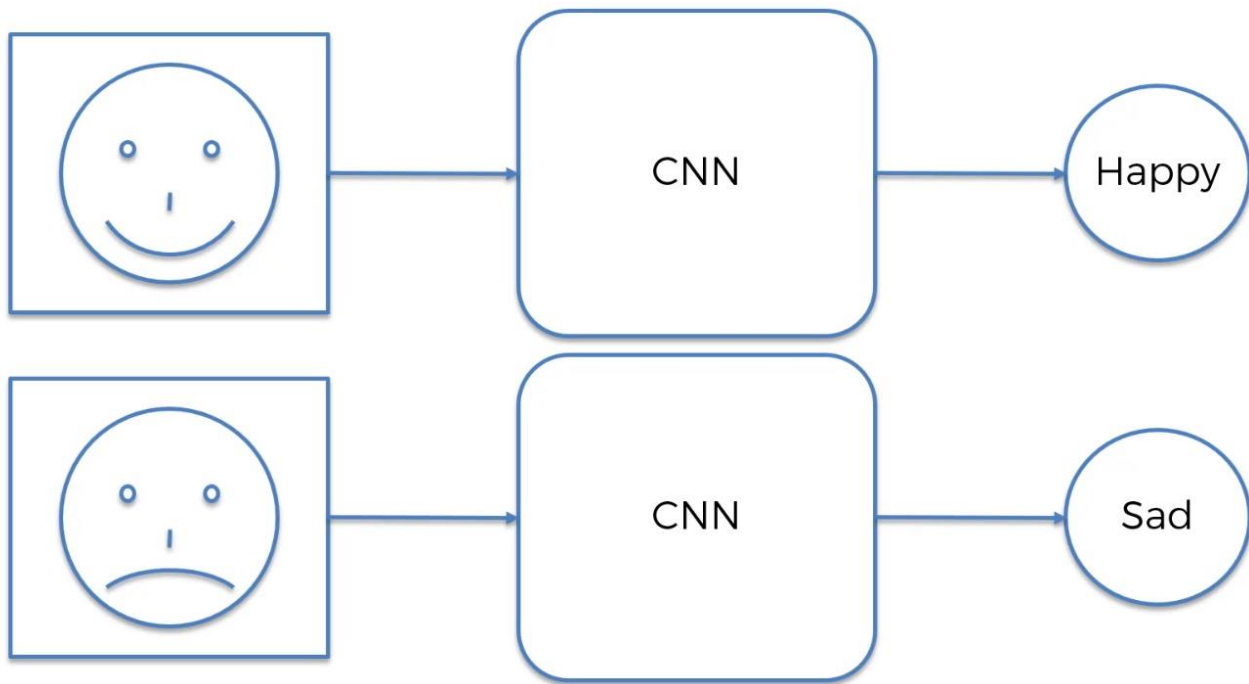
CNN



CNN

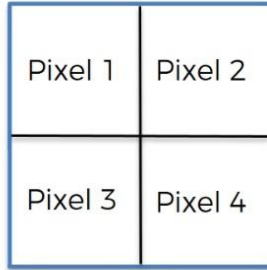


CNN

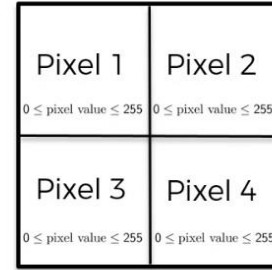


CNN

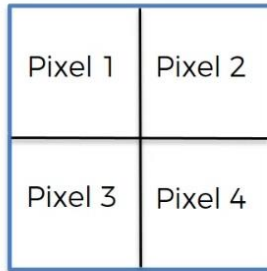
B / W Image 2x2px



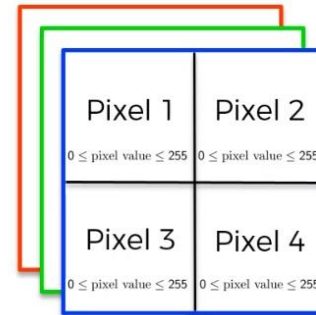
2d array



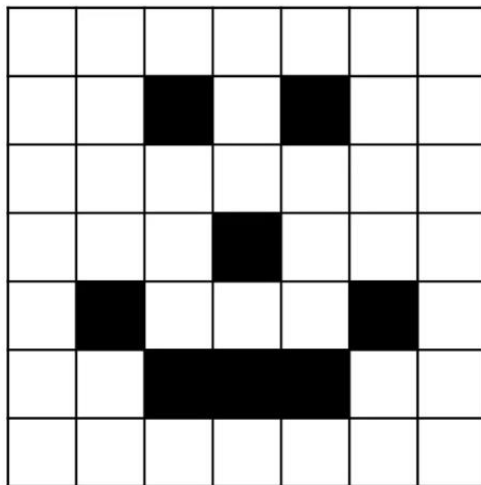
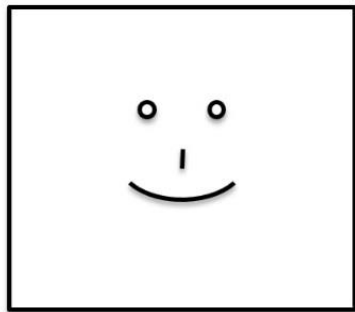
Colored Image 2x2px



3d array



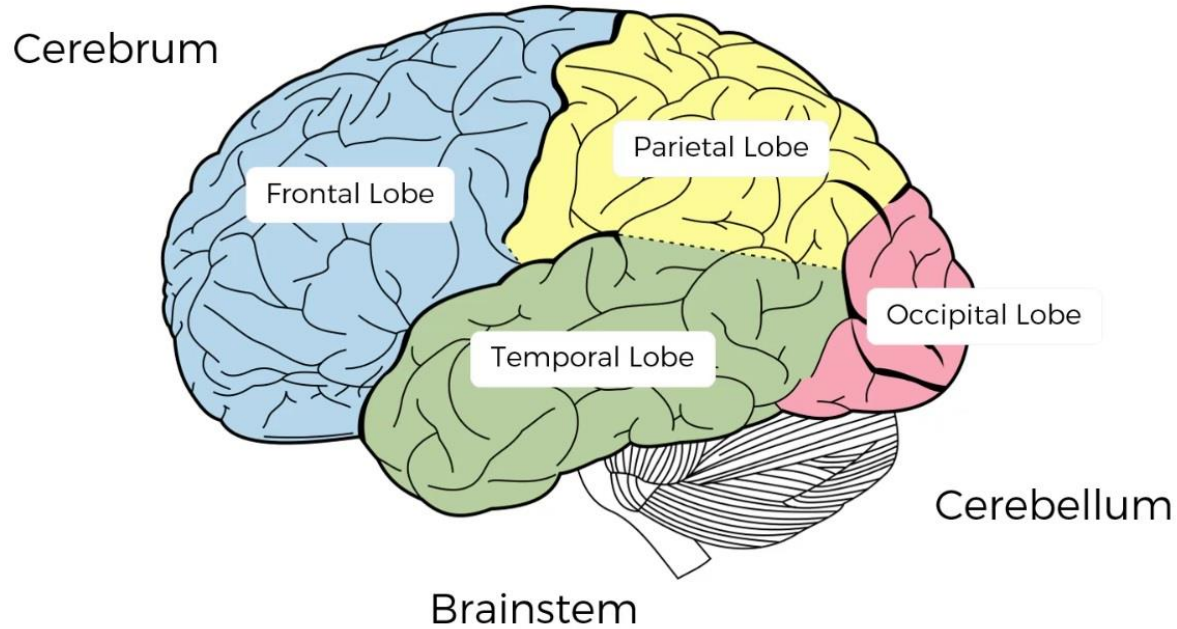
CNN



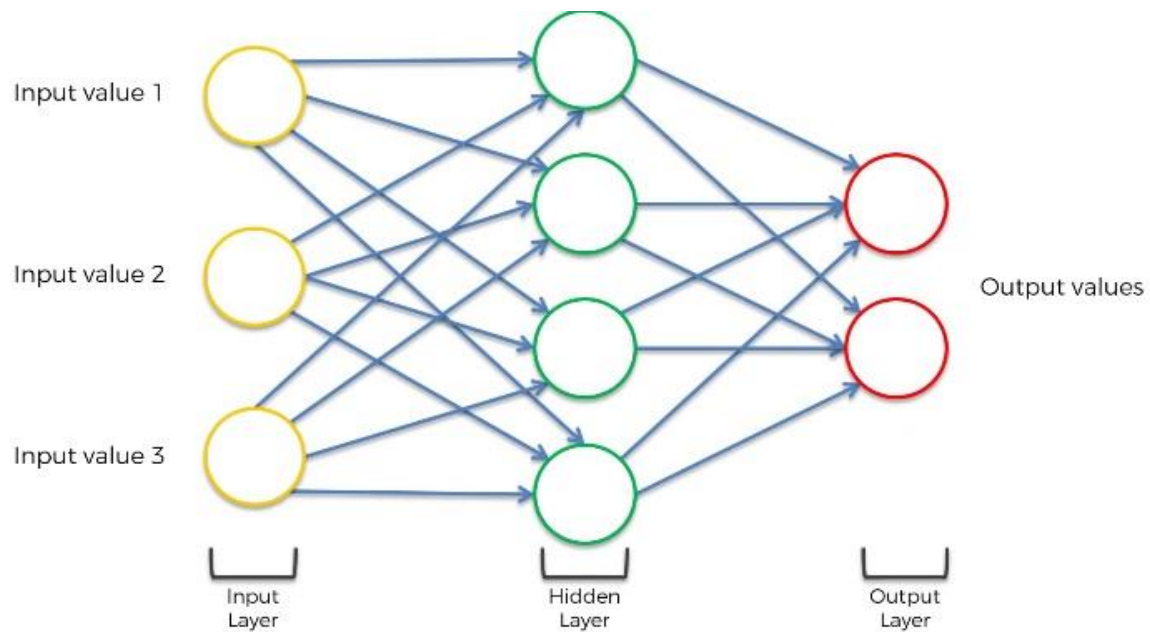
0	0	0	0	0	0	0
0	1	0	0	0	1	0
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	1	0	0	0	1	0
0	0	1	1	1	0	0
0	0	0	0	0	0	0

RNN: RECURRENT NEURAL NETWORKS

RNN



RNN

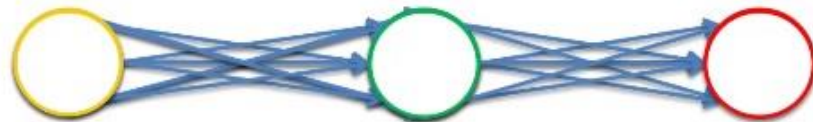


RNN

Input value 1

Input value 2

Input value 3



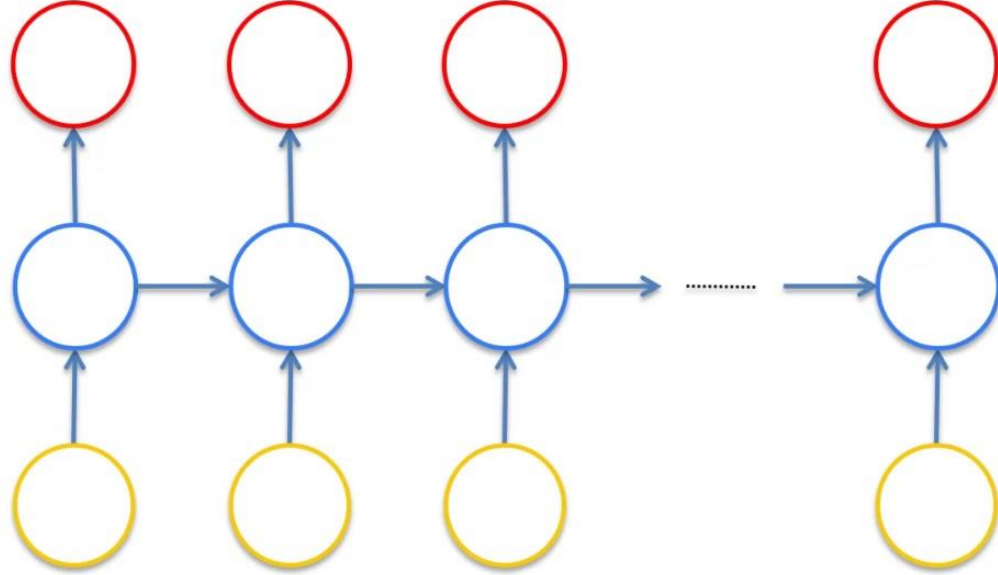
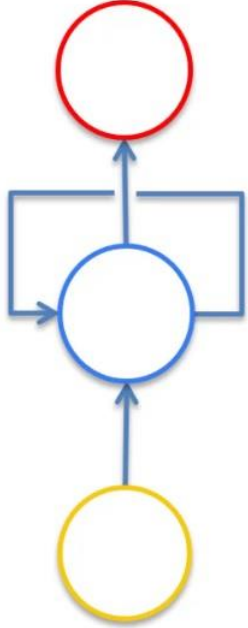
Output values

Input
Layer

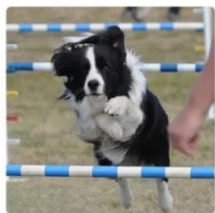
Hidden
Layer

Output
Layer

RNN

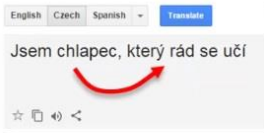


RNN EXAMPLES

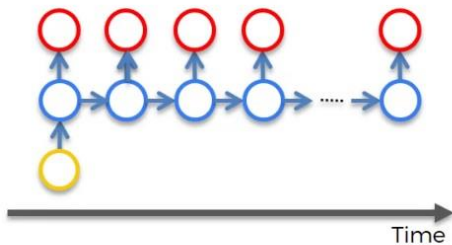


"black and white
dog jumps over
bar."

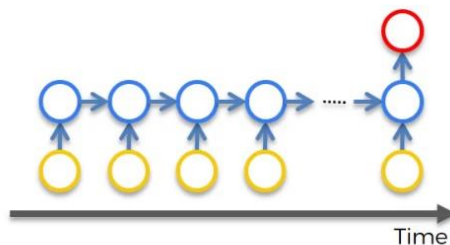
karpathy.github.io



One to Many



Many to One

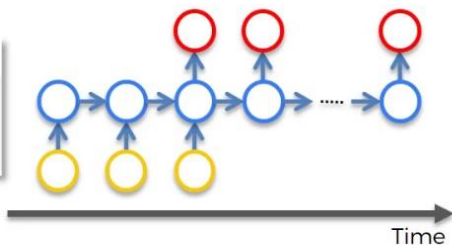


"Thanks for a great
party at the
weekend, we really
enjoyed it!"

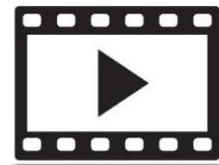
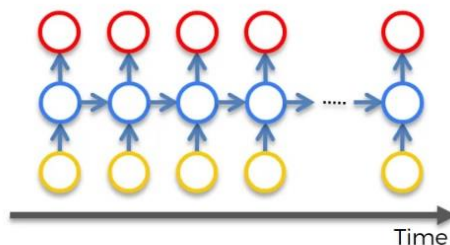
sentiment: positive
score: 86%

dev.havenondemand.com

Many to Many



Many to Many



Reference: *karpathy.github.io*

ANY QUESTIONS??

THANK YOU